

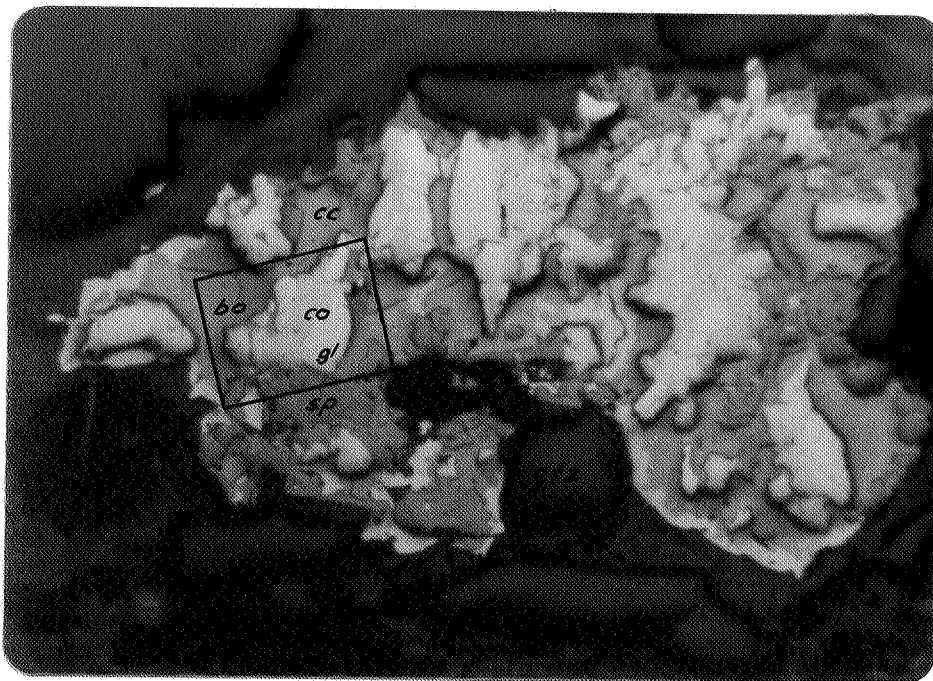


FIGURE 3

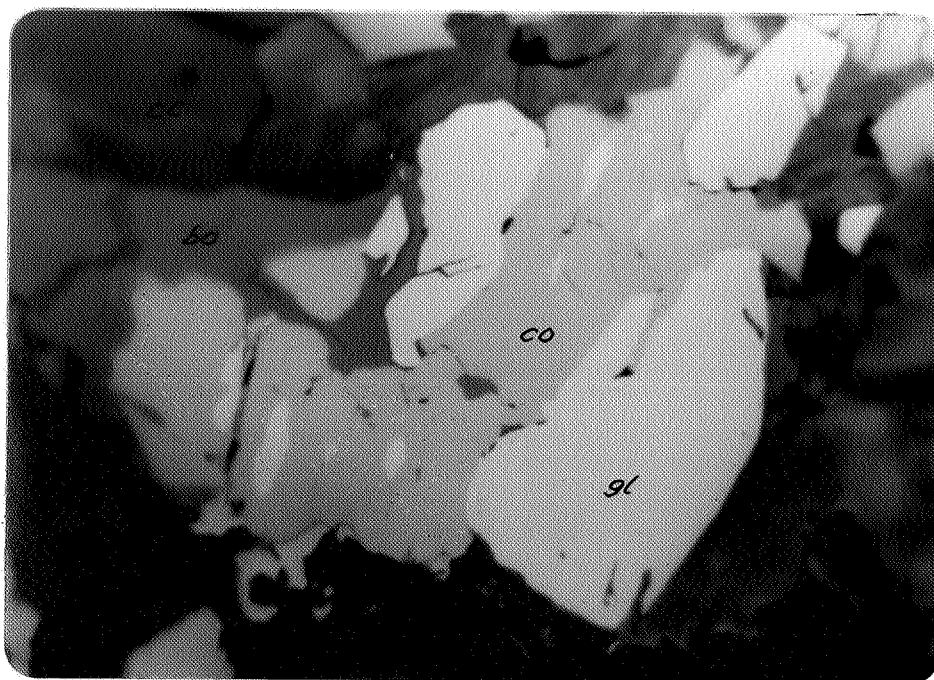
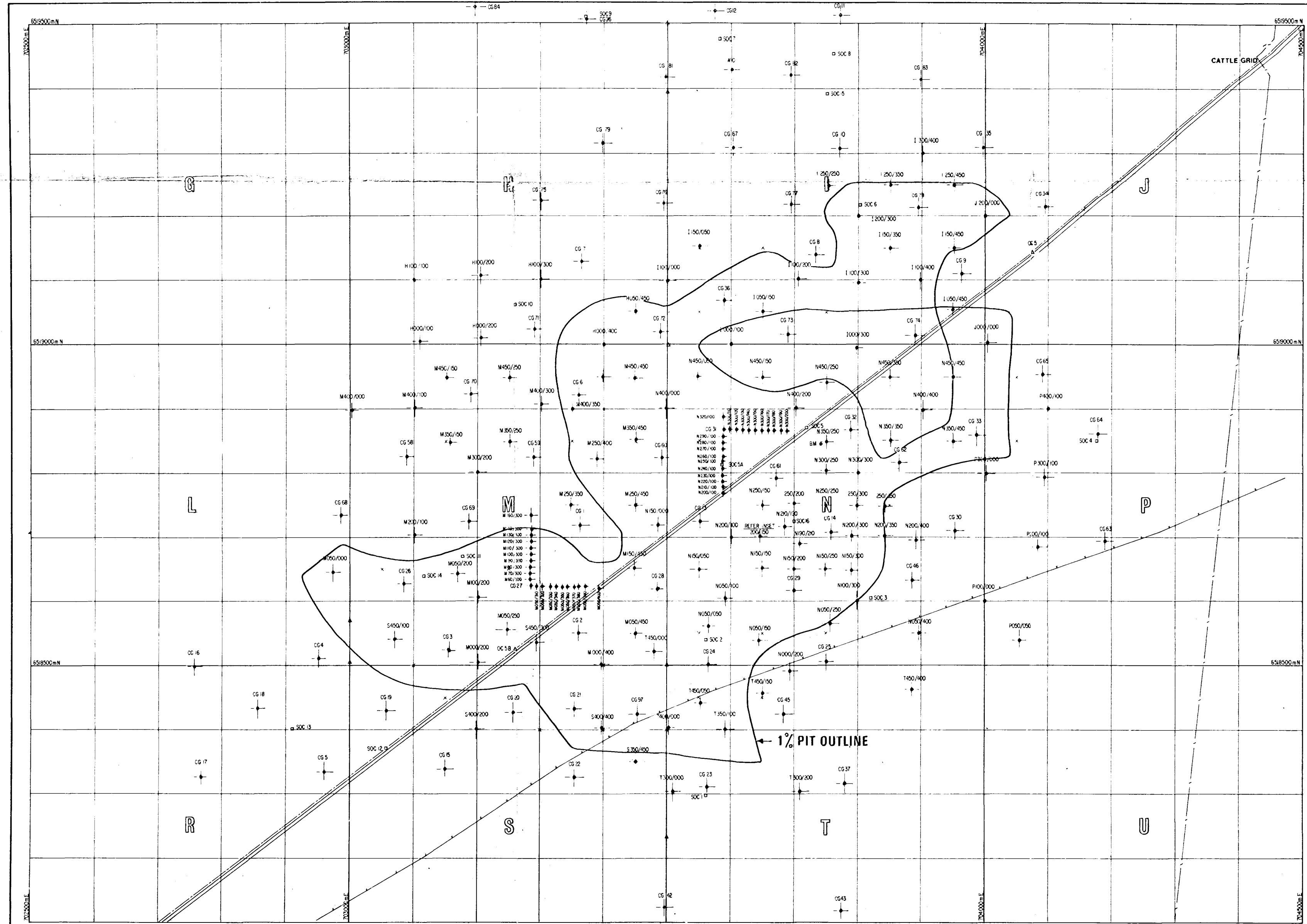
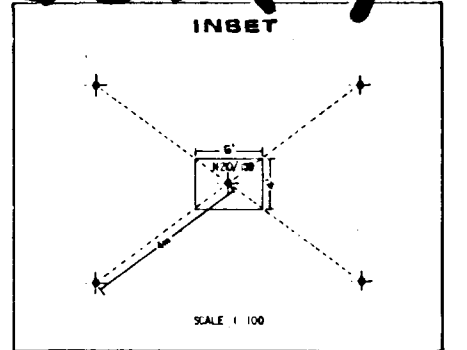


FIGURE 4



- REFERENCE
- Traverse
 - == Road
 - Power line
 - SOC Survey base
 - △ Survey datum peg
 - Drill hole
 - × Proposed drill hole
 - Fence
 - Drill hole, position approx

DRILLING LOCATION
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**CATTLE GRID AREA
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PACMINEX PTY LIMITED

FACIES VARIATIONS
IN THE WOOCALLA FORMATION
AND THEIR RELATION TO (THF)
BASE METAL MINERALISATION

PMR 58/74.

SYDNEY

May, 1974

K.J. Maiden

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PLAN

Dwg. No. 1525e: Distribution of Woocalla facies types.

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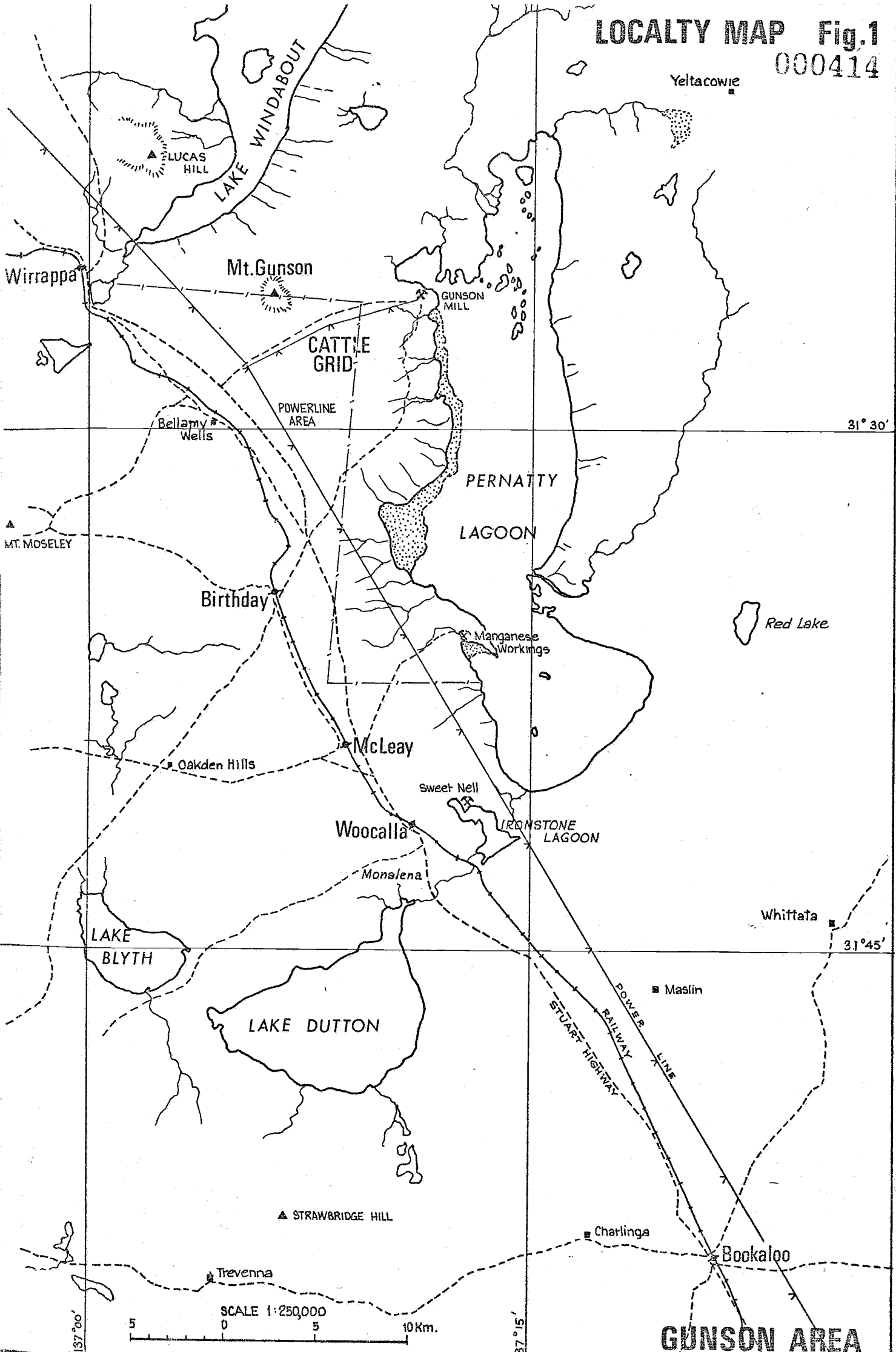
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KEYWORDS

MOUNT GUNSON
SH53-16 TORRENS
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COPPER
MINERAL DEPOSIT
EXPLORATION
METHOD
GEOLOGY
PETROGRAPHY

WOOCALLA DOLOMITE
SEDIMENTARY ROCKS
ENVIRONMENT
GENESIS
SULPHIDES
FACIES
GEOCHEMISTRY
ASSAY



1. INTRODUCTION

West of Lake Torrens, sediments of Upper Proterozoic age were deposited on the tectonically stable Stuart Shelf.

The Woocalla Dolomite lies disconformably on the Pandurra Formation, and is overlain by the Whyalla Sandstone. The unit crops out along the western shore of Pernatty Lagoon, at Sweet Nell, at Woocalla quarry and at the northern edge of Lake Dutton. In outcrop, the unit may be massive, laminated, brecciated or stromatolitic (Johns, 1968). In the Woomera Bore, the unit consists of 120 m of black shale. In the Winnie Pinnie-Trevenna area, Noranda drill holes intersected grey dolomitic shale and yellow to red clay (Appleyard, 1970).

With respect to base metal mineralisation, copper occurs at the Sweet Nell Mine in a shale unit underlying dolomite. High grade oxide copper ore was worked at this mine in the early years of this century. In addition, Noranda intersected minor Pb-Zn mineralisation (max. 0.63% Pb and 0.7% Zn) in dolomitic shale in the Winnie Pinnie-Trevenna area.

In the last few years, a large volume of data about the Woocalla Dolomite has accumulated. The aim of this report is threefold :-

- a) to summarise the available data;
- b) to relate the observed facies changes to the environment of deposition;
- c) to draw attention to the possibilities of economic base metal mineralisation in the Woocalla Dolomite.

2. SUMMARY

Pre-Woocalla erosion of the Pandurra Formation produced a surface of low ridges and stream channels leading into broad valleys. The sediments of the Woocalla Dolomite were deposited in a shallow sea on top of the post-Pandurra palaeo-surface. Geological mapping and stratigraphic drilling have shown that a variety of facies types can be recognised in the Woocalla Dolomite, and these can be related to the environment of deposition as follows :-

Deeper water	:	Black shale Red and green shale Dolomitic shale
Shallow water	:	Sandy dolomite Oolitic dolomite Dolomite breccia
Shallow lagoons	:	Laminated dolomite
Inter-tidal zone	:	Stromatolitic dolomite
Supratidal zone	:	Dolomitic mud and silt

Of these facies types, those deposited in deeper water, in shallow lagoons and in the supratidal zone may be associated with reducing conditions favourable for the precipitation of sulphides.

Data from recent stratigraphic drilling and from the Noranda exploration programme (carried out in 1968-71) indicate that high copper values are restricted to two main facies of the Woocalla Dolomite; the dolomitic mud and silt facies, and the dolomitic shale facies. Within these facies types, mineralisation appears to be related to local depressions in the post-Pandurra palaeo-surface.

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Concentrations of base metal sulphides have been intersected in drill holes in a number of areas, and in places the copper content exceeds 1%. The data indicate that there is a possibility of economic base metal grades occurring in the Woocalla Dolomite.

3. FACIES VARIATIONS IN WOOCALLA DOLOMITE

It is apparent from geological mapping and drilling that there are a number of significant facies changes within the Woocalla Dolomite. Although there are large gaps in our knowledge of facies types and distribution (due to lack of outcrop and difficulty in identifying rock type in drill cuttings), the following appear to be the main types. Numbers in brackets refer to samples examined microscopically by Mr. P.J. Curtis.

3.1 Black Shale (A9777)

This facies is a finely bedded black shale consisting of clays, carbonate and silica with minor graphite and sulphide. Graphite is considered to represent original organic material. Details of sedimentary structures are provided by Noranda drill logs.

In most places, black shale beds are separated by thin grey dolomitic beds, usually less than 1 cm thick and generally in the range 2-5 mm. The rock may be laminated, with black mudstone laminae alternating with grey carbonate laminae. In sections where there is no dolomite, the rock is more massive. Beds may be graded with black fissile mudstone at the base grading up to shaley dolomite at the top. Current bedding is recorded in several places but is not generally present. Scattered chips and flakes of dolomite and occasional thin sandy layers indicate minor intermittent influxes of coarser detrital material into the basin of deposition.

Bedding is normally horizontal but is disturbed here and there by structures resulting from slumping - wavy bedding, slumped bedding, pull-aparts, boudinage structures and minor brecciation. The upper surfaces of slump structures may be truncated to form micro-unconformities.

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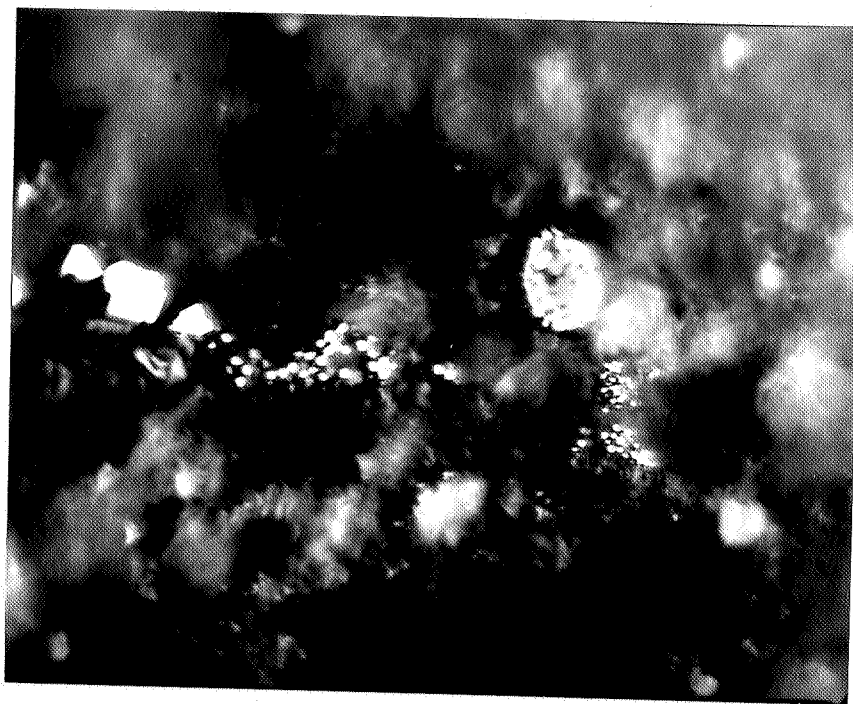


Figure 2: Pyrite framboids in black shale (Sample A9777 from 54 m depth in Hole OK8. Magnification x 980).

A number of terms used in the Noranda drill logs ("disrupted bedding", "smashed dolomite layers", "pile-up cascade structures", "diapiric structures") point to a certain amount of plastic injection of clay through the bedding prior to lithification.

Small scale faulting and one or two fault breccia zones are reported in the Noranda logs. Much more common are tension cracks, usually partly filled by calcite and sulphides. Occasional small calcite-sulphide lenses, apparently also formed by lateral secretion, are parallel to bedding. Calcite crystals and sulphides are also present in breccia zones.

Sulphides are visible in nearly all core intersections and drill cuttings. Curtis (1973) reported that the sulphide is mainly recrystallised framboids of pyrite (Fig. 2) which is considered to have been precipitated by organic processes. It occurs disseminated through the shale and in fine bands (up to 1 mm thick) parallel to bedding, as well as in lateral secretion veinlets. Other sulphides are galena, sphalerite and chalcopyrite, the last partly altered to chalcocite and covellite. They occur as disseminated grains, as blebs along bedding planes, as replacements of pyrite, and in lateral secretion veinlets. The sphalerite is a pale honey colour, indicating low iron content.

3.2 Dolomitic Shale

With increase in the proportion of carbonate laminae to mudstone laminae, the black shale grades into a dolomitic shale, which is usually grey to brown in colour, with occasional thin bands of black shale. The rock is laminated, and zones of slumping are reported in Noranda's drill holes LD5 and LD6 (Thomas and Douch, 1971). As for the black shale, this facies contains minor sulphide and graphite.

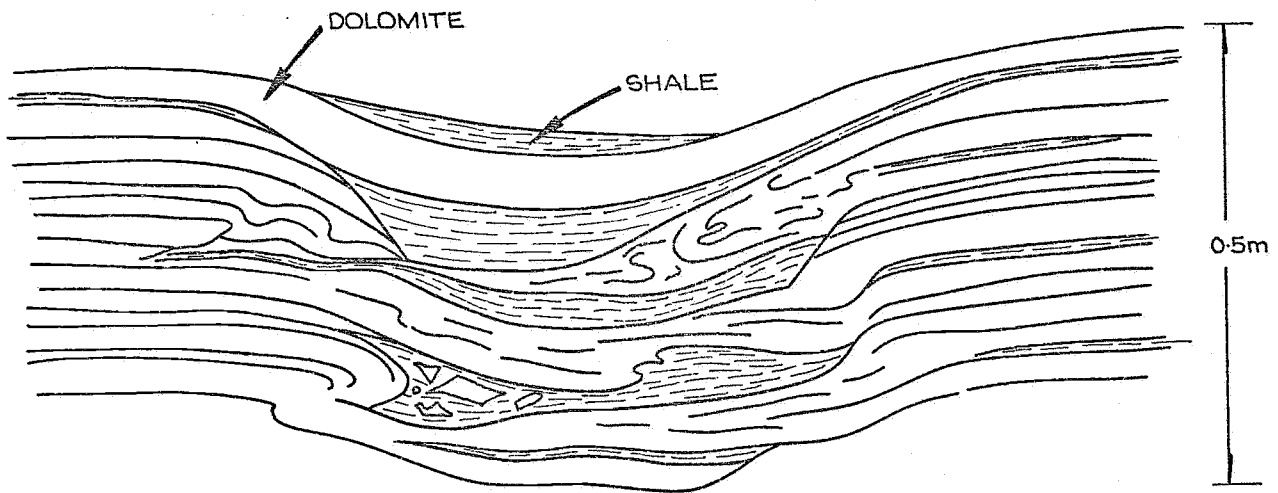


Figure 3. CHANNEL SECTION IN LAMINATED DOLOMITE IN WOOCALLA QUARRY

3.3 Laminated Dolomite

In the Woocalla Quarry, the rock is a pink to pale grey micritic dolomite, finely laminated, with secondary manganese oxide along lamination planes and in fractures. No sand grains, dolomitic mud clasts, oolites or algal structures are visible. The dolomite occurs in beds 10-200 mm thick, maintaining their thickness and regular spacing for distances of 20-40 m. The only bedding plane features are ripple marks with an amplitude of about 10 mm and a wave length of around 50 mm.

The dolomite beds are separated by narrow partings of white to pale yellow shale. Thickness of the shale is generally only 1-2 mm, but increases towards the top of the unit where shale beds of 20-50 mm thickness occur.

At irregular intervals the bedding is disturbed by channel structures consisting of slumped dolomite beds, shale lenses and some brecciated dolomite (see Fig. 3).

In the McLeay area, drill cuttings from holes OK20 and OK21 show that the Woocalla Dolomite here consists of a very fine grained, tough dolomite, finely laminated with white, green and pink bands. No sand grains, algal fragments, oolites or clay lenticles are seen in hand specimen. In this area, the laminated dolomite grades upwards into dolomitic shale.

3.4 Sandy, Oolitic and Breccia Dolomite (A9882, A9883)

These facies occur in close association in the field. Good outcrops are present along the western edge of Pernatty Lagoon, but in drill cuttings it is impossible to distinguish the various facies - the rock appears as grey to white crystalline dolomite, sometimes massive, sometimes bedded.

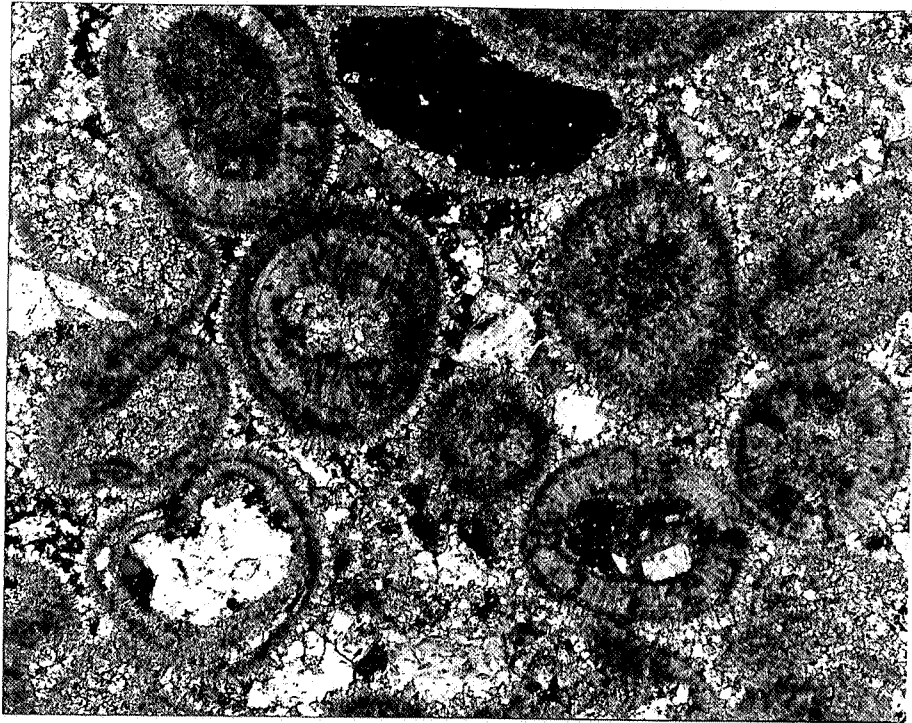


Figure 4: Oolitic dolomite (Sample A9882) from Pernatty Lagoon. Magnification x 90)



Figure 5: Stylolites disrupting Collenia structures. (Sample A4244 from Pernatty Lagoon. Magnification x 14).

Rocks consist of the following constituents in varying proportions :-

- (i) Oolites, with a core of crystalline dolomite, quartz or limonitic wad (Fig. 4).
- (ii) Algal oncolites
- (iii) Quartz sand grains
- (iv) A variety of transported dolomite fragments (stromatolitic and sandy varietites, mud flakes, etc.)
- (v) Matrix of recrystallised carbonate

The carbonate is (Ca, Mg, Mn) CO₃. In outcrop, the carbonate may be partly replaced by silica and limonite especially towards the base of the unit. Thin-sections show partial replacement of carbonate structures (oolites, etc.) by silica. Secondary manganese oxide is common. Post-depositional barite indicates saline conditions of deposition (Curtis, 1972). The occurrence of stylolites (Fig. 5) shows that extensive interstratal solution existed after deposition.

Beds range in thickness from a few centimetres to about a metre. Sedimentary structures include cross bedding, slumped bedding, minor brecciation, load casts, thin sandstone beds, ripple marks and transported dolomitic mud clasts.

3.5 Stromatolitic Dolomite (A9880)

Good exposures of stromatolitic beds (Fig. 6) are present along the western edge of Pernatty Lagoon. Beds are of the order of 20 cm thick.

In thin-section, the stromatolites are seen as wavy concentric structures of clear fine-grained carbonate, with associated recrystallised carbonate, manganese wad and some limonite. Between the structures are detrital fragments of dolomite, limonite and sometimes quartz and chert grains, in a matrix of fine dolomite.

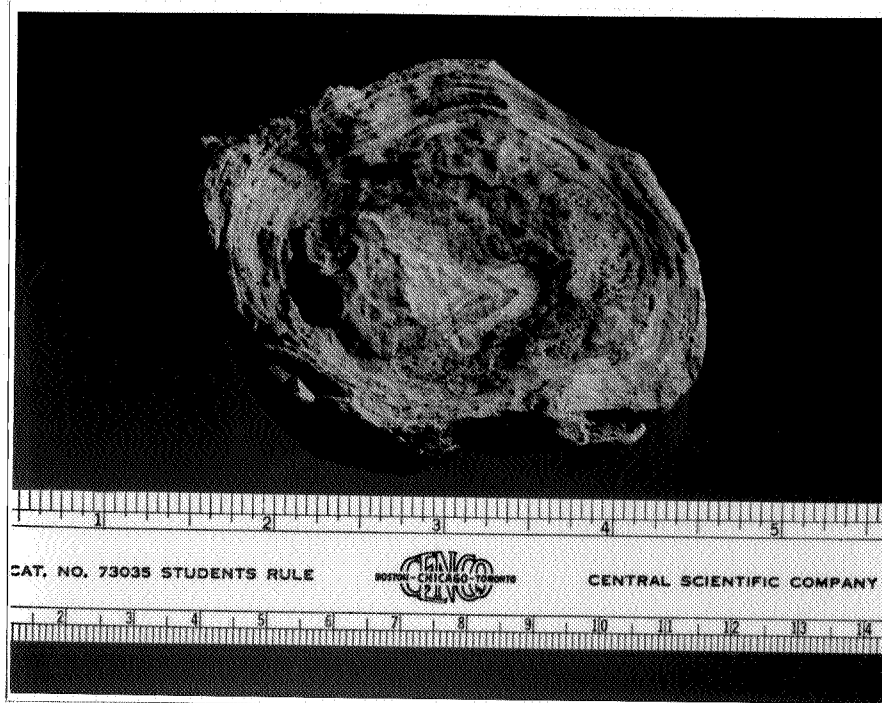


Figure 6: An example of a stromatolite (Pernatty Lagoon).

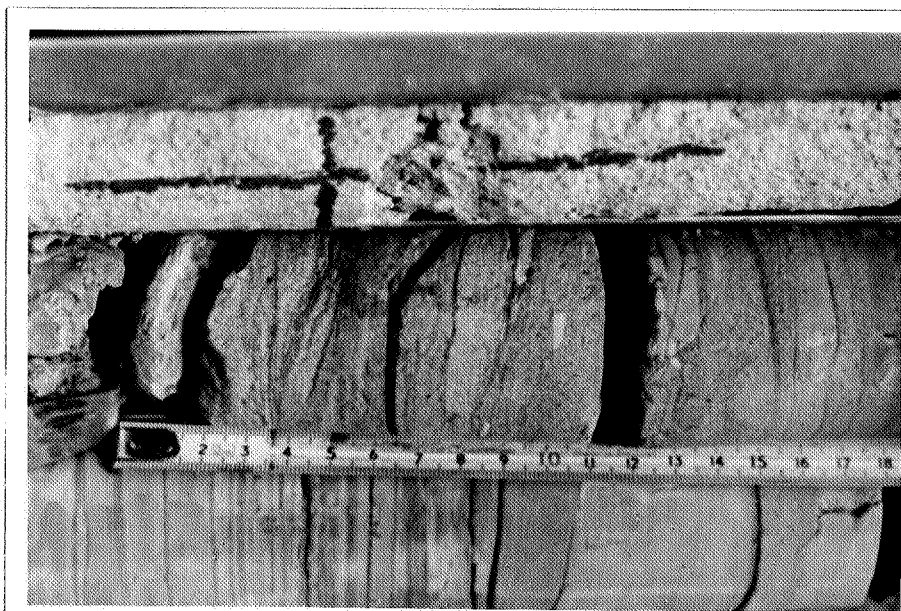


Figure 7: Dolomitic mud and silt with fine sandstone bands. (Hole MG14D).

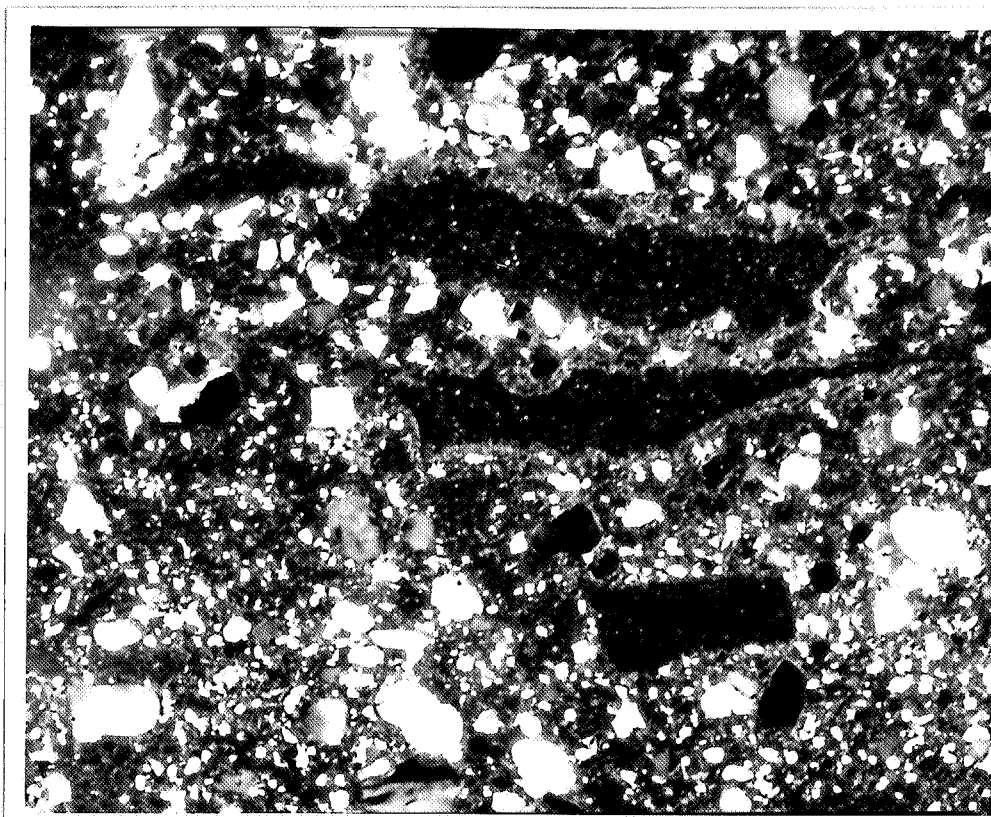


Figure 8: Dolomitic mud flakes in detrital matrix.
(Sample A9958 from Hole MG14D.
Magnification x 17).

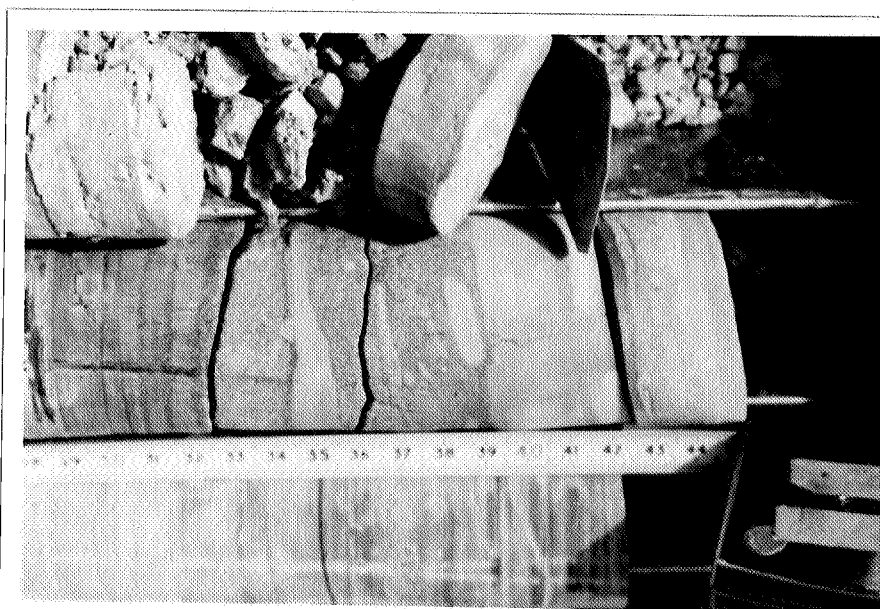


Figure 9: Intraformational breccia in dolomitic
mud and silt (Hole MG14D).

Stromatolite beds are sometimes separated by thin sandstone beds (A9879 and A9881) consisting of detrital grains of quartz and dolomite in a matrix of carbonate and silica. Scattered grains of limonitic manganese oxide are seen in one section (A9881).

3.6 Dolomitic Mud and Silt

This facies of the Woocalla Dolomite outcrops only in the vicinity of Sweet Nell (the so-called Sweet Nell Shale) but has been cored (in holes MG14D and LY3, 1 km N.E. of the Cattlegrid) and encountered in a number of rotary and percussion drill holes in the Bellamy Wells area (OK11), the McLeay area (OK20) and the Gully area (between the Main Open Cut and the Gunson camp).

In the MG14D core, the rock consists of dark mudstone bands interbedded with light grey to yellowish siltstone and fine sandstone bands (Fig. 7). Mudstone beds (A9959), consist mainly of fine-grained carbonate with minor quartz, illite and sulphide. Occasional transported quartz-illite and dolomitic mud flakes (Fig. 8) are present. Sandstone beds (A9960, A9961 and A9962) have detrital grains of carbonate, quartz and quartzite in a carbonate-quartz-illite matrix. Detrital carbonate grains are a mixture of inorganic dolomite and stromatolite fragments. Small grains of limonite and leucoxene were seen in some sections.

Interbedded with the mudstone and siltstone are bands of intraformational breccia (Fig. 9) consisting of fragments of fine-grained dolomite (usually oriented parallel to bedding) in a mudstone or sandstone matrix.

Beds are very thin, ranging from about 2 to 50 mm. Within each bed, a faint lamination may be present, and a trace of cross lamination is seen in some of the sandy beds.

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Figure 10: Cavities possibly due to solution of anhydrite or gypsum (Hole MG14D).

Other sedimentary structures include lensing out of beds, small lode casts at the base of sandy layers, dolomitic mud flakes, minor disconformities (truncated bedding planes), small clastic dykes and a possible "flame" structure indicating plastic injection of clay-rich material into sandy layers. Rectangular and triangular cavities (Fig. 10) in some beds may be due to solution of anhydrite.

Sulphide occurs as disseminated grains, particle aggregates, and in lateral secretion veinlets filling fine fractures. Apart from pyrite, the minerals carollite, chalcopyrite, bornite, digenite and covellite have been recognised. Much of the sulphide (in veinlets and attached to quartz grains) has recrystallised diagenetically, but some, disseminated through fine carbonate mud, appears to be syngenetic. The grain size of this material is very fine (around 0.06 mm). Pyrite occurs as framboids with a bornite and possibly greigite cement (A9963). Pyrite appears to displace the soft sulphide matrix of the framboids to form relatively large pyrite euhedra.

3.7 Red and Green Shale

Between "Pandurra" and "Hesso" the black shale facies of the Woocalla Dolomite grades into a red and green shale facies which apparently fills a fairly deep trough immediately to the east of the Pandurra Formation "high". Little is known of this rock type, and no petrographic examinations have been made.

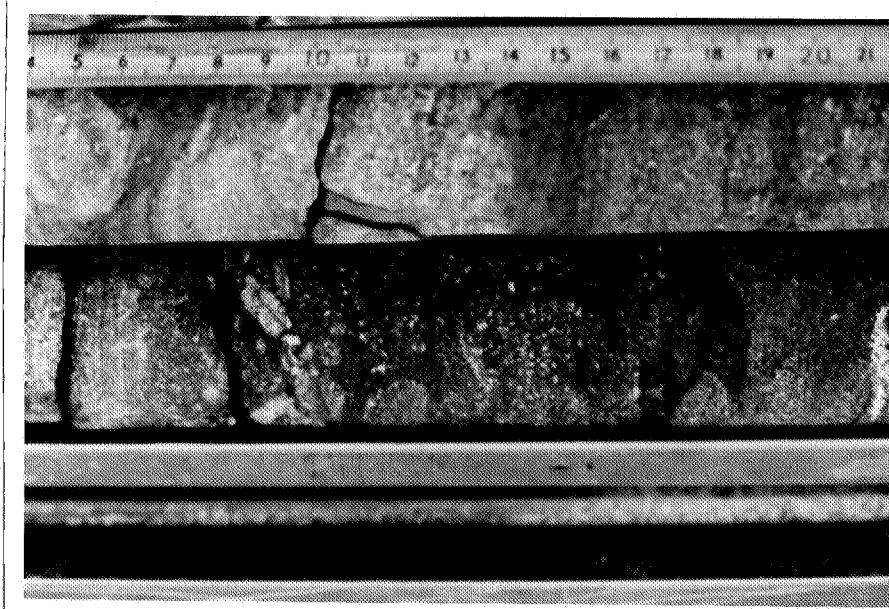


Figure 11: Pandurra regolith - a quartzite breccia
with a red clay and sand matrix.
(Hole MG14D).

4. PALAEOGEOGRAPHIC INTERPRETATION

4.1 Woocalla Sea

After lithification, the Pandurra Formation was uplifted and exposed to the agencies of erosion forming a surface of low ridges and broad valleys. Evidence for this can be seen on present day re-exposed Pandurra surfaces on the western side of Pernatty Lagoon. This erosion surface is marked by a quartzite breccia with a red clay and sand matrix (Fig. 11) which probably represents a fossil scree or soil horizon.

The main feature of the post-Pandurra palaeo-surface is a series of hills known as the Pernatty Culmination, extending northwards from "Yudnapinna" homestead to the northern end of Lake Windabout. Drilling and mapping have shown that the main post-Pandurra ridge on the western side of Pernatty Lagoon extends further westwards into the Oakden Hills area as a complex of low ridges and valleys. To the north, west and south of Oakden Hills, the post-Pandurra palaeo-surface slopes away to a broad shallow basin, rising again at Lake MacFarlane in the west and in the vicinity of Strawbridge Hill to the south.

The "McLeay Beds" were deposited as a thin veneer on the post-Pandurra palaeo-surface in scattered localities, but have little effect on the post-Pandurra topography.

During Woocalla time, the post-Pandurra erosion surface was largely inundated by a marine transgression, to form a shallow sea, partly enclosed by land. The various Pandurra highs represent islands in this sea - at Strawbridge Hill, Mt. Mosely, Lake Windabout and the main post-Pandurra ridge stretching along the western edge of Pernatty Lagoon (Fig. 12). To the east of the Pernatty Culmination, the Woocalla Sea apparently extended to Lake Torrens, but there is no data available for this area.

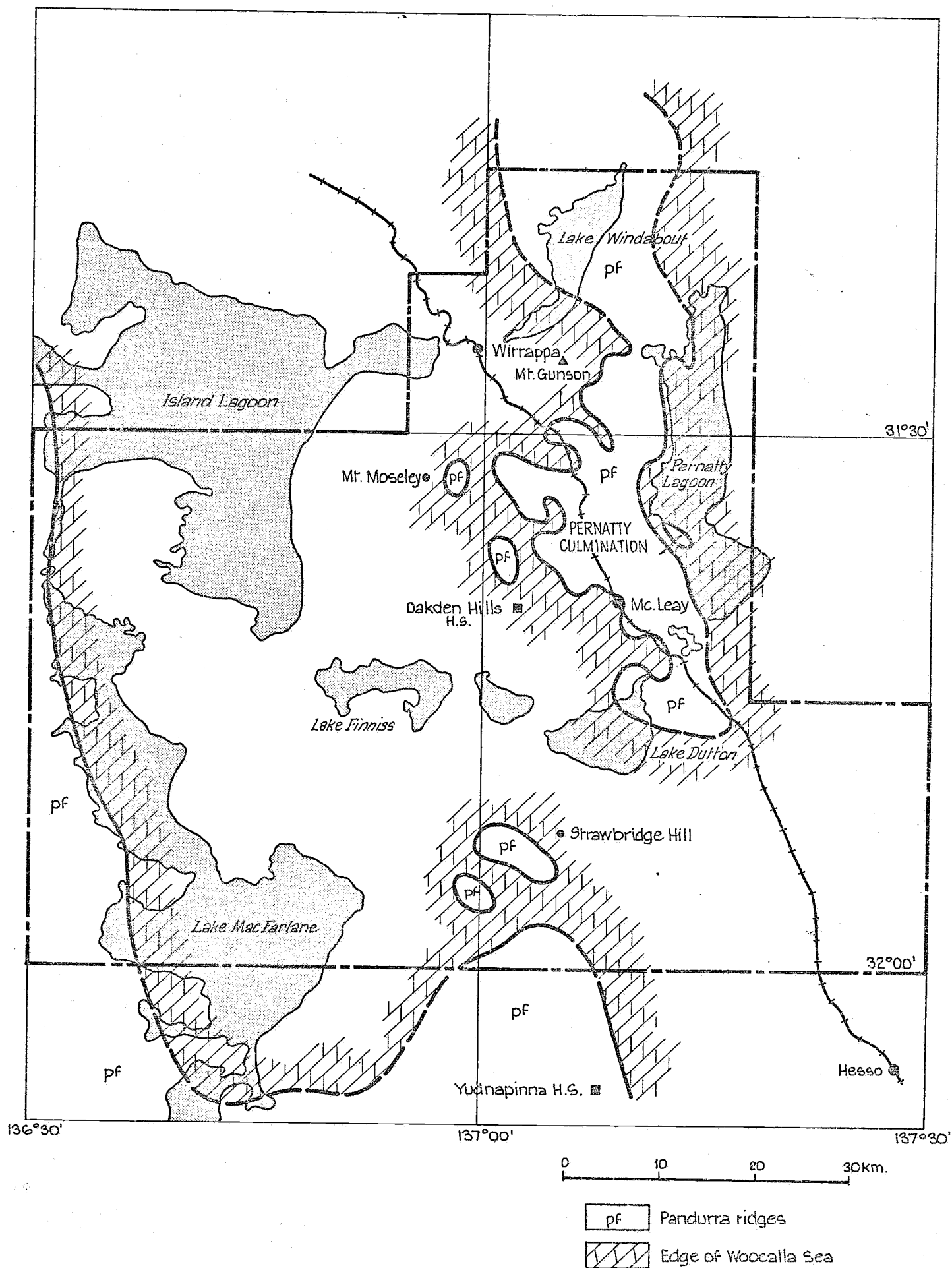


Figure 12. THE WOOCALLA SEA

4.2 Theoretical Pattern of Carbonate Deposition

"The general trend of increasing depth away from shoreline in most shallow seas affects waves and currents in a predictable fashion. In turn, differential action of waves and currents on the bottom affects grain size and sorting of the sediment, and causes formation of various sedimentary structures as well as influencing certain factors that control biotic distribution. As a result, a distinct and rather consistent pattern of sedimentary facies grading laterally into one another perpendicular to the shore is established." (Heckel, 1972).

Irwin (1966) described the theoretical model for non-detrital sedimentation in an epicontinental sea. He recognised three major zones characterised by different intensities of water agitation and resulting from the effect of different relative depths on the position at which waves and currents impinge on the sea bottom and dissipate their energy. Heckel (1972) compared Irwin's model to the actual facies changes in the sediments of the Persian Gulf (Fig. 13) which is a modern epicontinental sea with extensive carbonate sediments extending seawards from the Trucial Coast along the southern shore.

Zone I : In deeper water below effective wave base, the bottom is sufficiently undisturbed to allow fine sediment to settle out. Sedimentation may be very slow, for it is dependent solely on detritus derived from the adjacent shoreward zone for the source of mud. This zone of reduced deposition would also be a site of formation of authigenic minerals such as glauconite. Thus in the deeper water along the axis of the Persian Gulf, mud is accumulating. It is predominantly carbonate with a subordinate amount of shells and fine terrigenous material. Organic content of the sediment is high, and the resulting rock type will be dark shaley, shelly calcilutite.

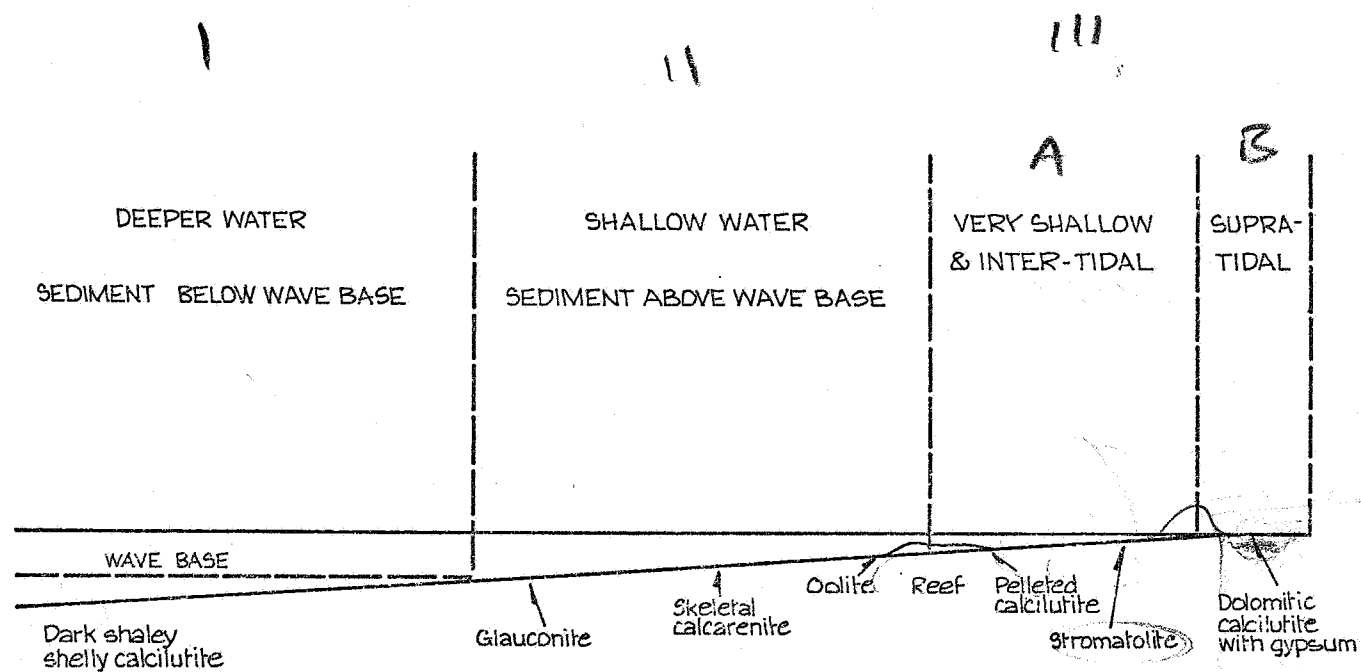


Figure 13. CARBONATE SEDIMENTATION ON SOUTH SIDE OF EPICONTINENTAL PERSIAN GULF

Zone II : In the next shoreward zone, effective wave base intersects the bottom surface and sediments are deposited in water agitated by waves. Little mud can settle out, so mainly sand-size and coarser material is deposited. In the absence of detrital influx, the material is largely skeletal debris, forming calcarenites. Cross-bedding and ripplemarks are characteristic. In the Persian Gulf, the deeper-water muds grade shoreward into skeletal calcarenite, with glauconite in the deeper offshore part of the facies.

Also in this zone reefs and oolite sands develop under certain conditions. In the Persian Gulf, along the shallowest fringe of the calcarenite zone are coral reefs and oolite shoals associated with tidal deltas and channel complexes.

Zone III : In the nearshore zone of shallowest water, the environment is again quiet because waves and currents are damped as their energy is dissipated on the shallowing sea bottom. Thus mud is again allowed to accumulate, its source in part being shoreward transport of the fine products of disintegration from the adjacent seaward zone. Lime mud may also be derived by direct physico-chemical precipitation from sea water. The resulting rock type is calcilutite, often with faecal pellets. If the climate is arid, evaporites may form in this zone. In the Persian Gulf, behind the reef area are very shallow lagoons with accumulations of lime mud containing a large amount of faecal pellets.

At the shoreline are the intertidal and supratidal environments. The intertidal zone consists of extensive flats covered with blue-green algal mats, as found in the Persian Gulf. The algal mats trap fine calcareous mud and silt, detrital

grains and tests of micro-organisms. These become covered to form soft laminated sediments under the algal sheath. Build-up of this material produces algal stromatolites. Between the stromatolite structures are calcareous detritus and sand grains.

Algal stromatolites are of two main types - Collenia, generally associated with a protected intertidal mud flat, and Cryptozoon, generally associated with an exposed intertidal headland (Logan, Rezak and Ginsburg, 1962). These types are distinguished from algal oncolites which are found typically in agitated lower intertidal conditions.

Intertidal sediments may be cut by tidal channels which become filled with carbonate mud, sand and coarser fragments.

The supratidal zone is covered by water only by spring or storm tides, and lies exposed sub-aerially for long periods of time. The sediments of this zone are described in some detail by Lucia (1972). They consist mainly of laminated carbonate mud containing fine grained dolomite, gypsum crystals and less commonly anhydrite and celestite. Alternating wetting and drying lithifies and tends to break the supratidal sediment into lithoclasts. Although they are indicative of the supratidal environment, such lithoclasts may be transported and thus deposited in other environment.

4.3 Comparison of Woocalla Sediments to Theoretical Pattern

A comparison of the Woocalla facies types to those of the theoretical model and of the modern sediments in the Persian Gulf, reveals many similarities. On the basis of these similarities it is possible to interpret the environment of deposition of the Woocalla Dolomite.

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Zone I : Deeper Water - Sediment below Wave Base

In the Woocalla Sea, the deeper parts of the basin of deposition are represented by the black shale facies, which grades shoreward into the dolomitic shale facies. The main differences between these and the Persian Gulf sediments are:-

- lack of shells
- lack of glauconite
- presence of large amounts of terrigenous material (clay and silica)

The thickness of the unit (up to 130 m) is also much more than expected due to influx of fine terrigenous material and perhaps gradual sinking of the sea floor.

Between "Pandurra" and "Hesso", the black shale changes shoreward into red and green shale. Cross sections show that this facies occupies a broad channel at the western edge of the Woocalla Basin (Fig. 14). This facies indicates a change to detrital deposition in an oxygenated environment as opposed to the reducing environment of the black shale facies.

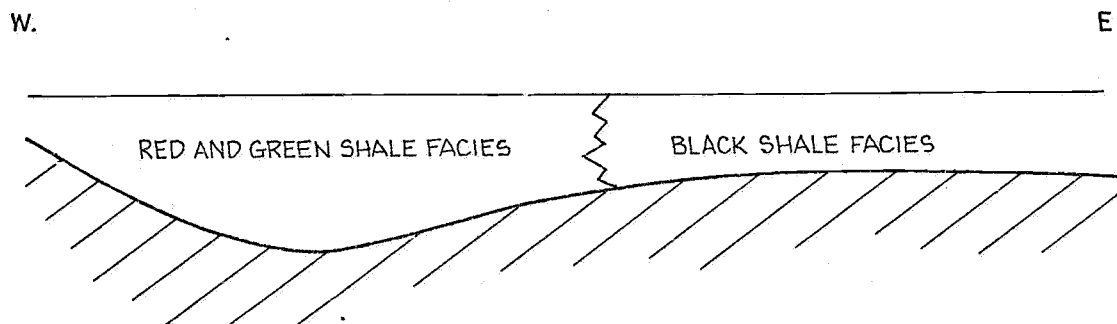


Fig. 14: Generalised section through Woocalla Dolomite in "Pandurra" to "Hesso" area.

000438

Zone II : Shallow Water - Sediment Above Wave Base

This zone is represented by cross-bedded sandy dolomite and oolitic dolomite. Brecciated varieties are the result of slumping or wave action. The differences between these and the Persian Gulf sediments are :-

- lack of skeletal material
- presence of detrital quartz grains
- lack of reef structures (although it is possible that undetected algal reefs may be present).

The thickness of sediments in this zone is variable. Up to 25 m have been intersected in drill holes in Pernatty Lagoon.

Zone III : Very Shallow Water - Sediment Not Affected by Waves

Three of the Woocalla facies apparently belong in this zone.

- a) Laminated Dolomite - this facies may be equivalent to the pelleted calcilutite found in the Persian Gulf. If so, it would have formed by accumulation of lime mud in shallow lagoons, protected from wave and current activity by reefs (if they existed) or sand banks.
- b) Stromatolitic Dolomite - this is a distinctive inter-tidal facies formed in mud flats. Columnar stromatolites, of the type found along the edge of Pernatty Lagoon, are abundant in Proterozoic sediments but are rare in modern environments; the only known modern analogs to these ancient forms are found at Shark Bay, Western Australia. (Hofmann, 1973).

The thin sandstone interbeds indicate periods of greater current activity, either by tides, storms or a general rise in sea level.

c) Dolomitic mud and silt - this facies probably represents the supra-tidal zone, as indicated by the following features: -

- fine laminations *Bank ORP*
- dolomitic mud flakes *main, SPP DAM, Bellamy wells*
- cavities possibly due to solution of gypsum
- thin sandstone beds (material brought in by storm tides). *LW60*

The presence of fine quartz and illite again shows a mixture of terrigenous material with the carbonate sediment.

4.4 Reducing Environments

Reducing environments occur where the lack of atmospheric oxygen allows organic matter to be preserved. Partial anaerobic decay of accumulated organic debris produces methane and a number of other largely unknown organic substances. Sulphur from original protein is converted to hydrogen sulphide. Some sulphur combines with iron to form iron sulphides, which eventually crystallise as pyrite.

In shallow seas, such as the Woocalla Sea, reducing environments may occur :-

- (i) In deeper water, below the zone affected by waves and currents, i.e. the black shale and dolomitic shale facies.
- (ii) In shallow lagoons protected from wave and current activity by reefs or sand banks, i.e. the laminated dolomite facies.

- (iii) In the supratidal zone, beyond the reach of normal tides, i.e. the dolomitic mud and silt facies.

4.5 Distribution of Dolomite Facies

On the eastern side of the Pernatty Culmination the Woocalla sea-floor sloped away gently, and stromatolite beds were deposited on extensive tidal flats in the vicinity of central Pernatty Lagoon and the Manganese Workings. At the Manganese Workings, the stromatolitic dolomite is underlain by dolomitic mud and silt facies. Towards the northern end of Pernatty Lagoon, sandy, oolitic and breccia dolomites were deposited close inshore. In the Gully area (between the Main Open Cut and the Gunson camp) dolomitic mud and silt was deposited first, then overlain by sandy and stromatolitic dolomite.

On the western side of the Pernatty Culmination, the Woocalla sea-floor sloped away quite steeply into deeper water. In the Mt. Gunson, McLeay and Oakden Hills area, the shoreline fringe of dolomite is generally less than 500 m in width before grading through dolomitic shale into black shale. A number of shallow inlets stretched back towards the main Pandurra Formation ridge, and dolomitic mud and silt were deposited in shallow supratidal lagoons. Sediments deposited in these inlets have been intersected in drill holes northeast of the Cattlegrid (Hole MG14), in the Powerline area (PL1), in the vicinity of Bellamy Wells (OK11) and near Birthday Siding (OK35 and OK36). The Sweet Nell shale is an outlier of dolomitic mud and silt which was originally connected to the Woocalla Sea either in the vicinity of Woocalla quarry or south of Ironstone Lagoon.

Dwg. No. 1525e is a generalised map of Woocalla facies distribution in the Mt. Gunson - Oakden Hills - Pernatty Lagoon area.

000441

W.

E.

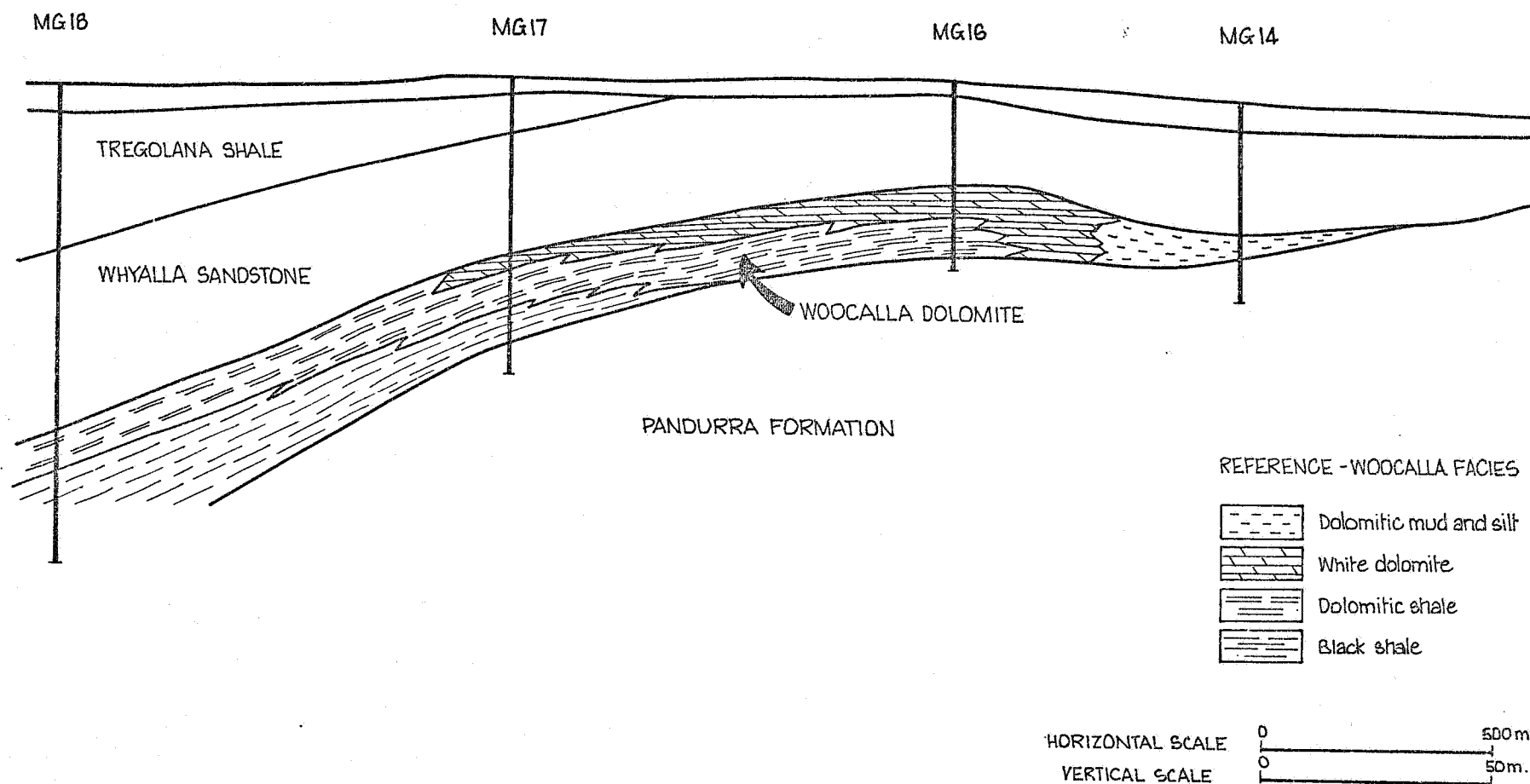


Figure 15 VERTICAL FACIES CHANGES IN DOLOMITE NORTH OF CATTLE GRID

In the Mt. Gunson area there is a vertical facies change from deeper water sediments up into shallow water sediments. For example, hole MG17 shows the following sequence :-

Depth -	40 - 42 m	:	White dolomite
	42 - 45 m	:	Blue-grey dolomite
	45 - 51 m	:	Dolomitic shale
	51 - 59 m	:	Black shale

This indicates a gradual marine regression through the latter part of Woocalla time (see Fig. 15).

In the Pandurra area, the Persian Gulf-type model is complicated by a greater influx of detrital material into the basin of deposition. Most of this material appears to have been derived from the Gawler Range volcanics, which must have formed a topographic "high" immediately to the west of the Pandurra area.

The dolomites here have a higher clay content than they do further north, so that in many places the rock is a grey dolomitic claystone. In addition, dolomite is interbedded with sandstone and even conglomerate bands, consisting of grains and pebbles of acid volcanics in a matrix of clay and carbonate.

000443

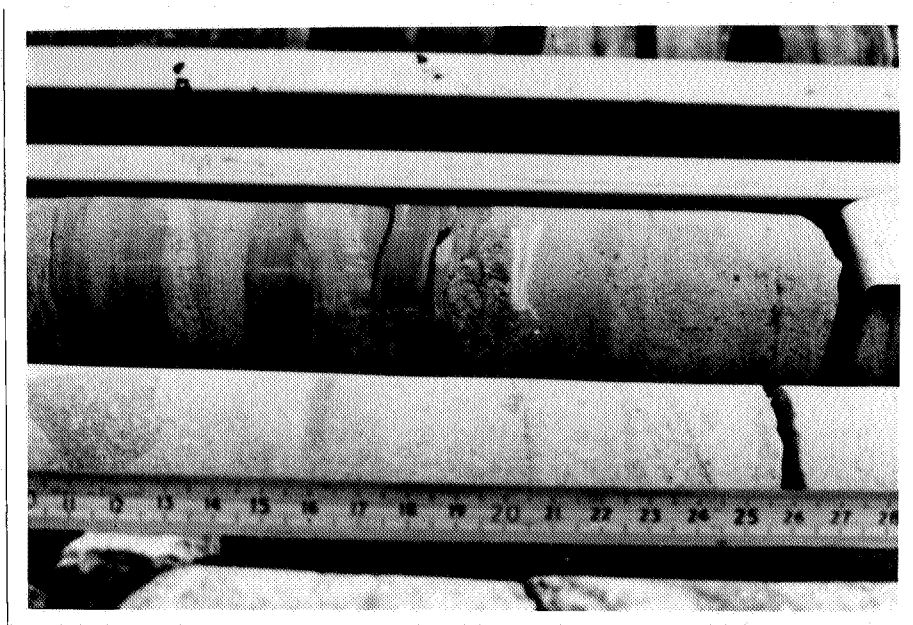


Figure 16: Basal shale with sandy layers.
(Hole MG14D).

5. BASE OF WOOCALLA DOLOMITE

5.1 Basal Breccia and Conglomerate

In many places the base of the Woocalla Dolomite is marked by a breccia or conglomerate consisting of fragments of dolomite and clay, and fragments or pebbles of Pandurra quartzite in a carbonate matrix. Ferruginisation and silicification are pronounced in outcrops along the western edge of Pernatty Lagoon. On an island in the southern part of Pernatty Lagoon, the basal Woocalla is a conglomerate consisting of red quartzite cobbles in a white quartzite matrix. In thin-section, the quartzite matrix is seen to be silicified dolomite. (Curtis, 1973A).

Noranda drill logs (Thomas and Douch, 1971) show black shale with lenticles of quartz sandstone (reworked Pandurra grains) and angular fragments of quartzite in the bottom half metre of Woocalla Dolomite in drill holes LD1, LD2 and LD3.

In part of the Pandurra area, the Woocalla Dolomite directly overlies the Gawler Range Volcanics. The basal Woocalla here is a conglomerate containing pebbles and cobbles of acid volcanics in a sandstone matrix with a dolomite cement.

5.2 Basal Shale and Sandstone

In the MG14 area (1 km northeast of Cattlegird) and in outcrops along the western edge of Pernatty Lagoon, the dolomite is underlain by green and red shale with thin sandstone beds.

In hole MG14D, dolomitic mud and silt grades down into shale (A9964) consisting of quartz, kaolinite, illite and carbonate, with minor limonite and leucoxene. Inter-laminated sandy layers (A9965) contain detrital quartz,

HOLE
MG 14D

HOLE
LY 3

000445

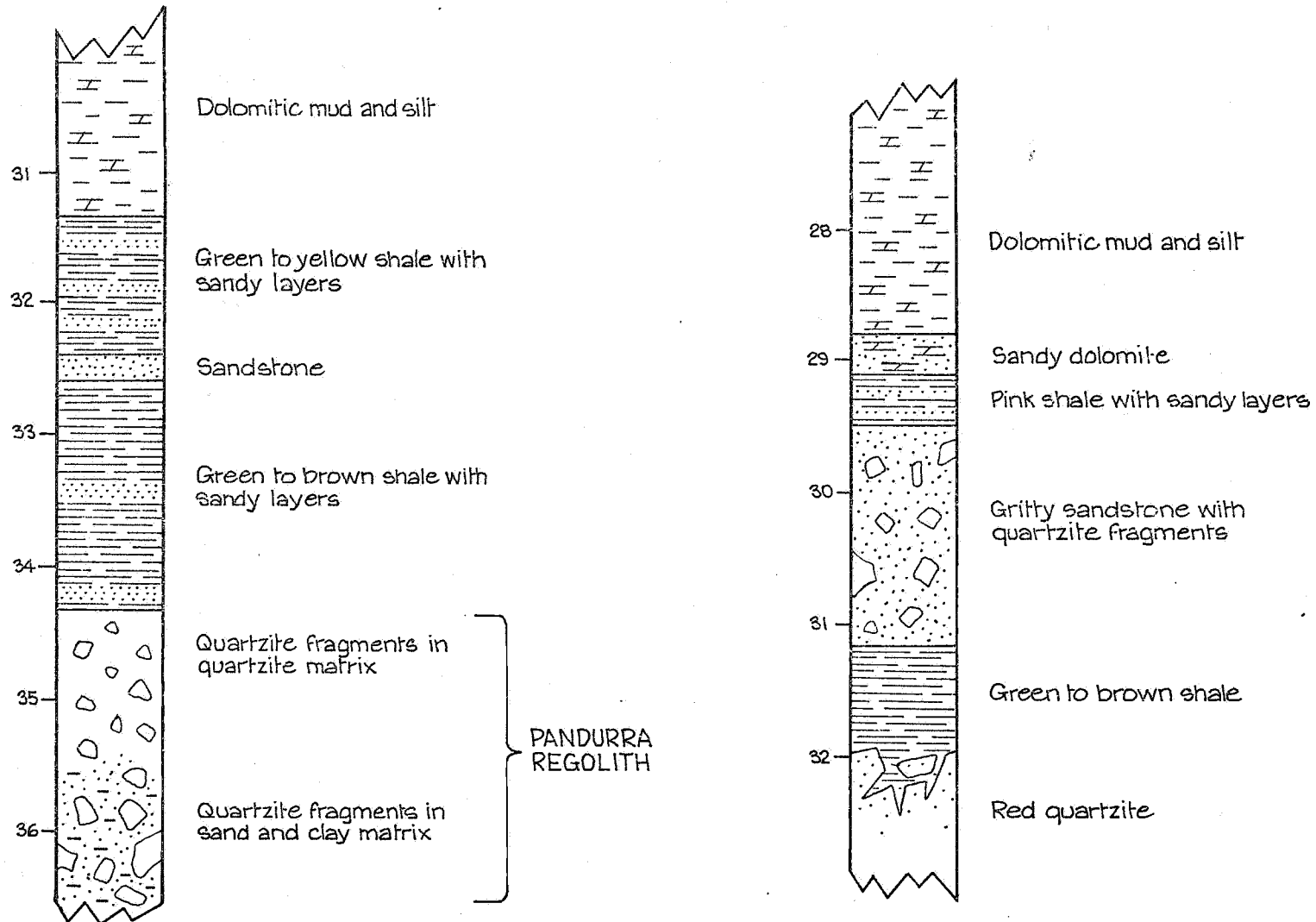


Figure 17. BASAL SHALE AND SANDSTONE IN MG 14 AREA

000446

chert, clay clasts and dolomite clasts in a matrix of carbonate, illite and limonite (see Fig. 16). In hole LY3 there is a bed of quartz grit consisting of quartz grains and red quartzite pebbles in matrix of soft sand and clay (see Fig. 17).

Shale (A9864) from the edge of Pernatty Lagoon consists of detrital quartz and chert grains and muscovite flakes in a quartz-limonite matrix. Interbedded sandstones have detrital quartz and chert in a kaolinite-illite-limonite matrix. The limonite component is quite noticeable in the central Pernatty Lagoon area. The basal Woocalla member is readily traced by the ferruginous outcrops.

5.3 McLeay Beds

This name has been given to a sequence of rocks disconformably overlying the Pandurra Formation in the vicinity of McLeay railway siding (see Fig. 18). Capping the red quartzite of the Pandurra Formation is a ferruginous deposit with pebbles and fragments of quartzite; this is the regolith marking the post-Pandurra palaeo-surface. Above this are 8 - 10 m of quartzite, silicified shale, sandstone and conglomerate. These rocks consist mainly of quartz (reworked Pandurra?) and chert grains in a matrix of chert, limonite and clay. Secondary silicification is extensive (Curtis, 1974B).

Geological mapping and stratigraphic drilling have identified similar rock types in a number of localities in E.L. 81. These observations suggest that the McLeay Beds are equivalent to the basal Woocalla.

In the Powerline area (7 km south of the Cattlegrid area) hole PL1 intersected 19 m of conglomerate, red sandstone and green and red shale grading up into Woocalla dolomitic mud. The shale in this locality is similar to the basal shale beneath dolomite in the MG14 area.

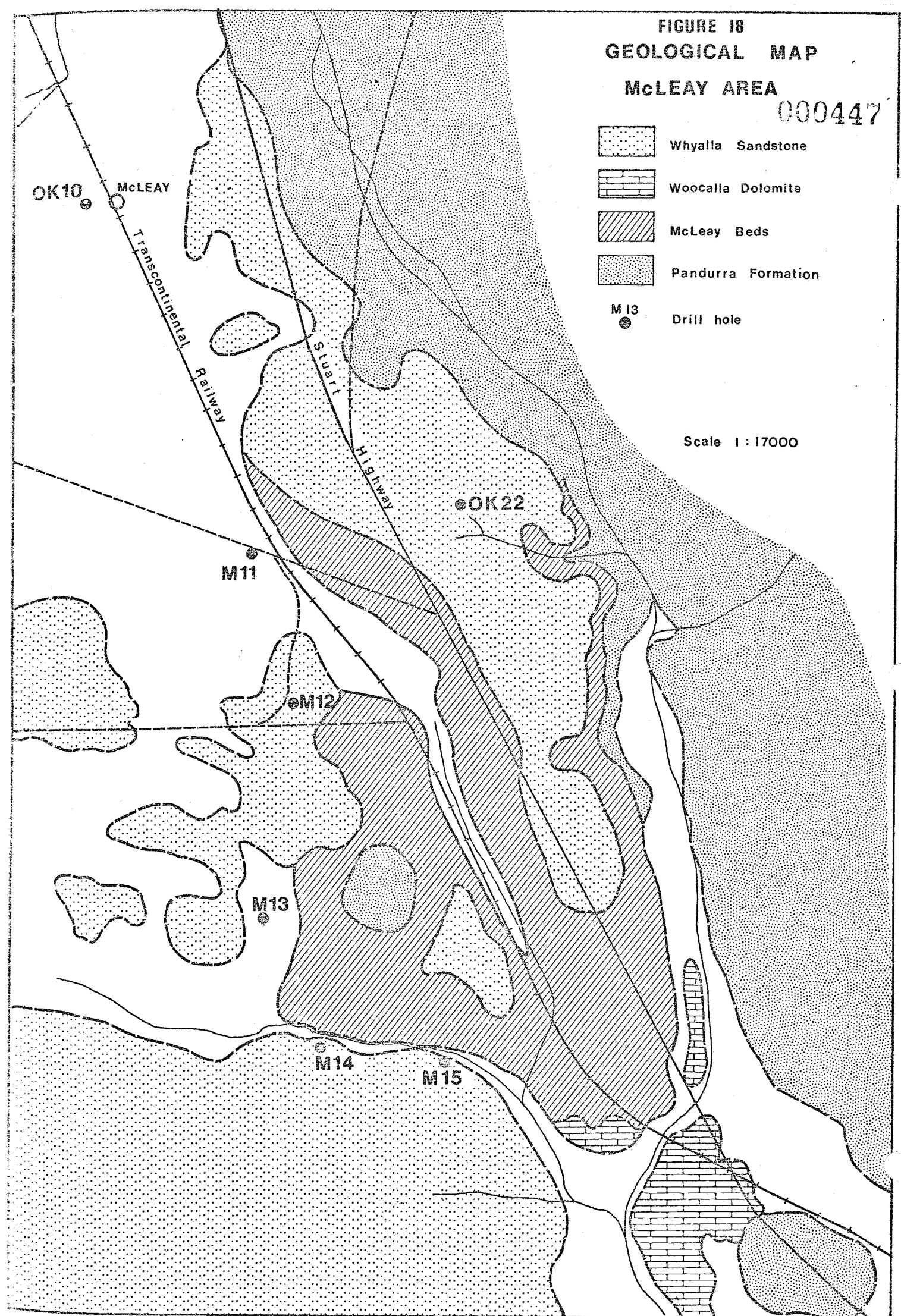
FIGURE 18
GEOLOGICAL MAP

McLEAY AREA

000447

-  Whyalla Sandstone
-  Woocalla Dolomite
-  McLeay Beds
-  Pandurra Formation
-  M 13 Drill hole

Scale 1 : 17000



000448

In the vicinity of the Manganese Workings, sandstone, siltstone and silicified shale underlie the dolomite. One sample (A9872) of white siltstone contains possible glass shards and cusped quartz grains, suggesting a tuffaceous horizon. (Curtis, 1973C).

An examination of core from the Woomera Bore has shown that the top 15 m of the Pandurra Formation is quite unlike the normal Pandurra Quartzite. It is a laminated grey to pink quartzite with scattered quartz pebbles. This intersection is tentatively correlated with the McLeay Beds (Appendix 1).

In the Oakden Hills area, the McLeay Beds consist of up to 9 m of interbedded quartzite, shale, sandstone and conglomerate, as intersected in drill holes OK41, OK42 and OK45.

The most distinctive rock type is the conglomerate described initially from outcrops along the Elizabeth River northeast of Mt. Gunson. The rock occurs sporadically along the main ridge of the Pernatty Culmination. It contains rounded pebbles of white quartz, quartzite, chert and red jasper in a siliceous matrix.

The McLeay Beds are not continuous but form a thin veneer on the post-Pandurra palaeo-surface in scattered localities. They appear to be largely reworked Pandurra material with some influx of foreign detritus, deposited in a continental or shallow marine environment. In the vicinity of the Manganese Workings and the Powerline area, the McLeay Beds appear to grade up into the Woocalla Dolomite. In the McLeay and Oakden Hills area, they may be lateral equivalents of the basal Woocalla, representing a continental facies.

5.4 Pandurra Gravel

In the Pandurra area, a gravel horizon underlies the Woocalla Dolomite. This unit contains pebbles and cobbles of a wide variety of rock type - Pandurra Quartzite, Gawler Range porphyry, quartz, jaspillite and basement metamorphics. The pebbles occur in a soft ferruginous matrix. The gravel is generally unconsolidated.

A remarkable feature is the thickness of this gravel unit. Hole EX80 intersected 54 m of it beneath Woocalla clay and overlying Pandurra Formation. Hole EX81 bottomed in gravel after intersecting 32 m of it beneath black shale.

This gravel unit is possibly equivalent to the McLeay Beds.

000450

6. TOP OF WOOCALLA DOLOMITE

6.1 Clay Unit

In the Oakden Hills-Lake MacFarlane area, soft grey clay, varying in thickness from 1 to 10 m, overlies the black shale. The nature of this material has not been determined, but it appears to be a mixture of mud and silt size particles with a certain amount of carbonate. This material has moderately high base metal values.

<u>Drill Hole</u>	<u>Clay Thickness</u> (m)	<u>Maximum Metal Conc. (ppm)</u>		
		<u>Cu</u>	<u>Pb</u>	<u>Zn</u>
OK8	1	75	130	260
MF2	10	60	70	210
MF4	10	250	120	180
MF5	3	100	300	200

Table I: Metal concentrations in grey clay in Lake MacFarlane area.

In the Winnie Pinnie-Trevenna area, Noranda drill logs show a sequence of buff, yellow and red clays (up to 10 m thick) overlying dolomitic shale. Thomas (1969) considers that the multicoloured clays and irregularities in the upper surface of the dolomite indicate a transition to arid conditions and some erosion.

The Magazine Hill clay which occurs as an isolated outlier in the Woocalla area, is now correlated with the clays at the top of the Woocalla Dolomite (Johns, 1972).

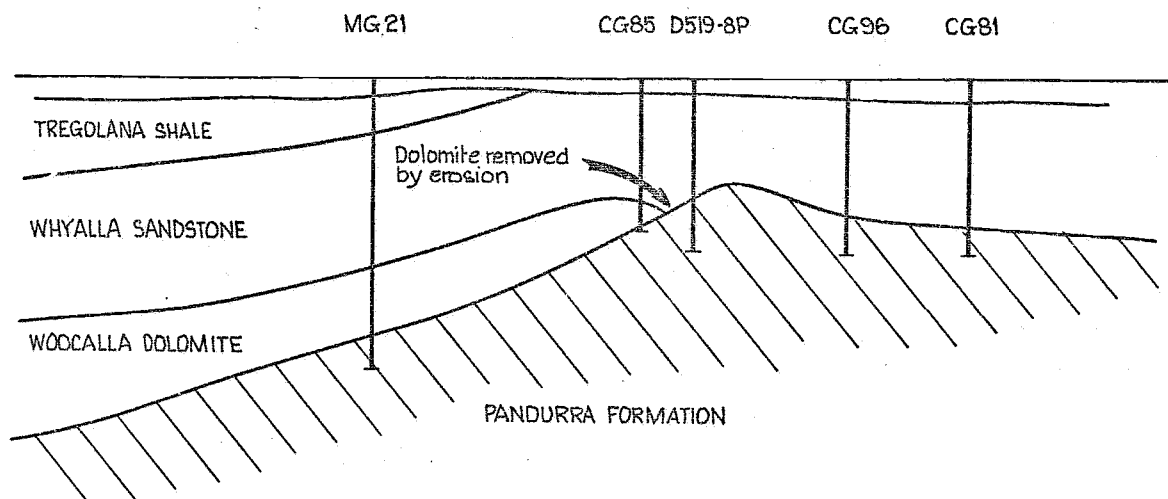
6.2 Woocalla-Whyalla Boundary

In the vicinity of the Pernatty Culmination, the Whyalla Sandstone rests on the Woocalla Dolomite at a distinct disconformity. The sediment changes suddenly from dolomite to sandstone with a breccia at the base of the sandstone. In the McLeay and Cattlegrid areas the

CATTLE GRID AREA

N.W.

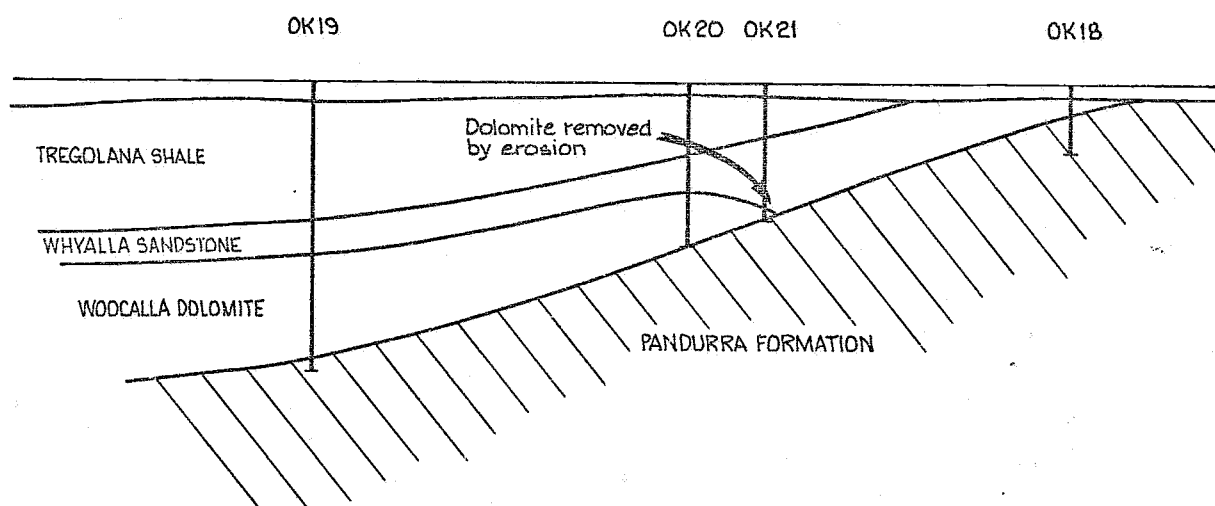
S.E.



MC. LEAY AREA

S.

N.



HORIZONTAL SCALE 1:10,000 0 200 400m.

VERTICAL SCALE 1:2000 0 40 80m.

Figure 19 CROSS-SECTIONS SHOWING EROSION OF EDGE OF DOLOMITE

edge of the Woocalla Dolomite, where the unit laps onto the Pandurra Formation, has been eroded away and Whyalla Sandstone fills an erosion channel (see Fig. 19). Along the western shore of Pernatty Lagoon, there is even a slight angular unconformity between the two units. The relationships imply that there was a time break after deposition of the Woocalla Dolomite, during which the dolomite was uplifted to some extent and partly eroded.

Away from the palaeo-shoreline, however, the two units are quite conformable and even grade into one another. In the Oakden Hills area the clay content in Whyalla Sandstone increases markedly towards the base of the unit, and in places the rock becomes a sandy claystone. In addition, carbonate cement and occasionally dolomite lenticles appear.

In the Lake MacFarlane drill holes, grey dolomitic clay at the top of the Woocalla Dolomite grades up into white sandy clay at the base of the Whyalla Sandstone.

Drilling by Noranda in the Lake Dutton area showed that the base of the Whyalla Sandstone is a shaley sandstone, up to 20 m thick, which is considered to be a transition into the underlying black shale. The drill log from hole LD2 describes the transition zone as consisting of thinly bedded layers of :-

- white to grey shale (containing black shale flakes)
- sandstone
- sandy shale
- occasional bands of black and grey-black shale

The conclusion drawn is that, towards the end of Woocalla time, the newly-deposited dolomite was being eroded in the vicinity of the Pernatty Culmination (i.e. at the shore of the Woocalla Sea), but deposition of sediment continued in the offshore areas. This period also saw a change from non-clastic carbonate deposition to fine clastic deposition.

7. BASE METAL MINERALISATION IN WOOCALLA DOLOMITE

7.1 Sweet Nell-type Mineralisation

Copper mineralisation at the Sweet Nell Mine occurs in a shale bed, approximately 1-2 m thick, underlying dolomite. The shale lies directly on the post-Pandurra palaeo-surface. The main sulphides are pyrite and chalcopyrite. Percussion drilling has outlined an area containing an estimated 213,000 tonnes of 1.22% Cu.

The Sweet Nell shale is an outlier of dolomitic mud and silt which was originally deposited in a shallow inlet, connected to the Woocalla Sea, either south of Ironstone Lagoon or in the vicinity of Woocalla quarry. Holes drilled into a shallow basin of dolomitic mud and silt at Woocalla intersected up to 0.22% Cu.

Other areas where the dolomitic mud and silt facies was intersected are:-

- a) Manganese Workings - up to 0.2% Cu in 1.5 m intersections.
- b) Gully area - values in the range 0.1 - 0.4% Cu over an area of almost 1 km².
- c) N.E. of Cattlegrid - Hole MG14D had values in the range of 0.2 - 0.7% Cu over a vertical interval of 4 m; one ½ m intersection assayed 2.53% Cu.
- d) Bellamy Wells Area - Hole OK11 had one value of 0.43% Cu in a 3 m intersection.
- e) McLeay Area - Hole OK20 had one 3 m intersection assaying 0.22% Cu; other drill holes showed that dolomitic mud and silt facies has a limited distribution in the McLeay area.
- f) Powerline Area - Hole PL1 intersected 8 m of dark-coloured dolomitic mud; assays are not yet available.
- g) Birthday Area - Holes OK35 and OK36 intersected 4-5 m of dolomitic mud; Cu values are 0.34% and 0.11% respectively.

000454

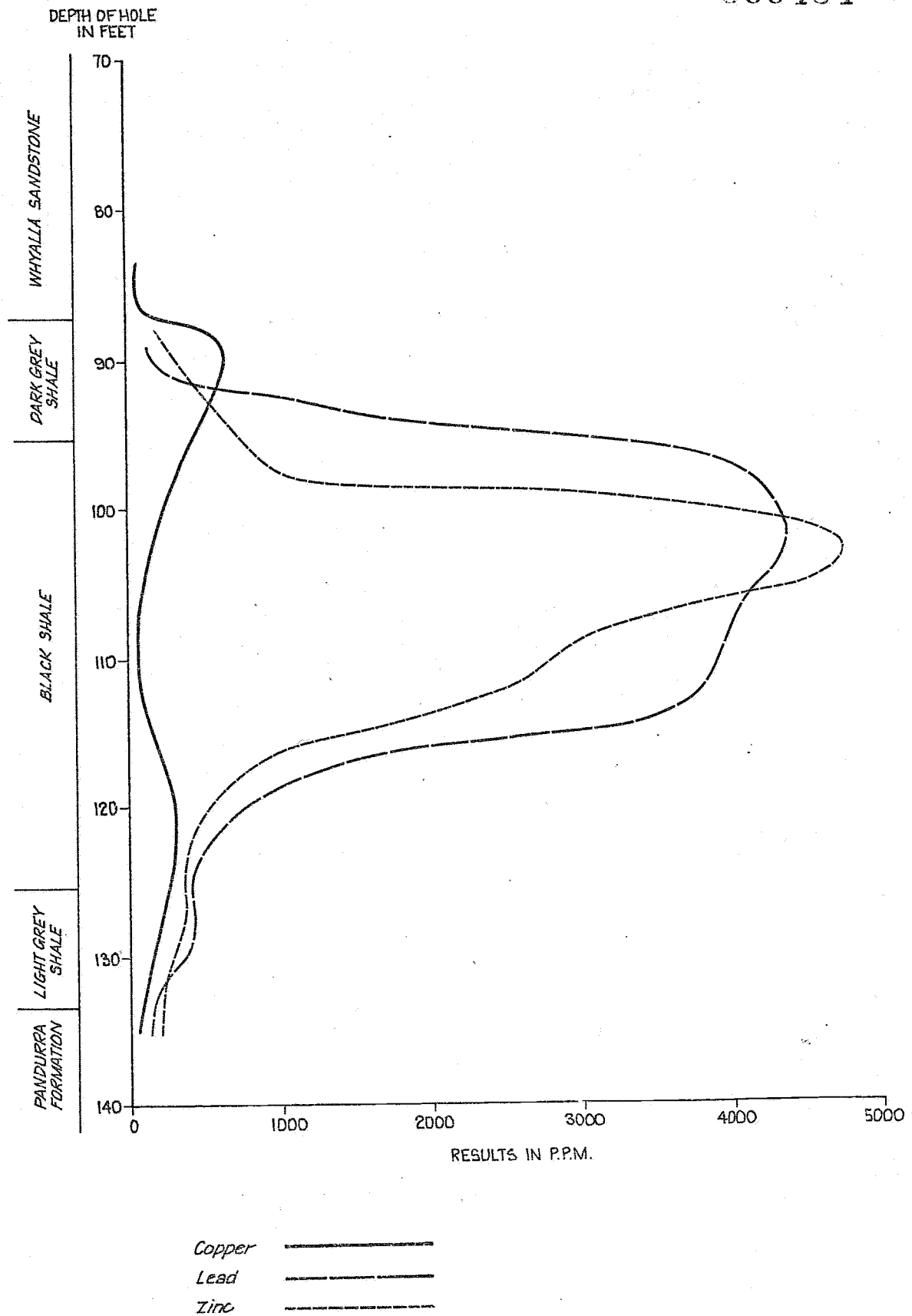


Figure 20. BASAL METAL DISTRIBUTION IN WOODCALLA SEDIMENTS IN WINNIE-PINNIE AREA

(AFTER NORANDA AUSTRALIA LIMITED)

The above figures show that there is a possibility of economic copper grades occurring in the Sweet Nell-type environment. Copper concentrations of 0.2-0.4% are normal for the dolomitic mud and silt facies, and may increase to over 1% in some areas.

7.2 Mineralisation in Black Shale and Dolomitic Shale

In a percussion drilling programme in the Winnie Pinnie-Trevenna area (south of Lake Dutton), Noranda delineated two zones containing sparse lead and zinc sulphides in black clay and dolomitic shale. The two zones trend north as shallow depressions in the post-Pandurra palaeo-surface. Shales over the more elevated portions of this surface are relatively deficient in base metals (Thomas, 1969).

The zones are both 6-9 m thick. One of the zones averages 0.2% Pb and 0.3% Zn over an area of 1.5 km²; the other averages 0.1% Pb and 0.1% Zn over an area of 2.5 km². Maximum values recorded are 0.63% Pb and 0.7% Zn over a 1.5 m intersection. Lead and zinc concentrations vary sympathetically. Copper values range up to 0.3%.

Overlying the black shale is a dark grey shale containing above normal copper values (see Fig. 20). Subsequently, the black shale was cored in 6 drill holes spaced about 10 km apart. Minor amounts of sulphide and above normal metal values were detected in every hole. The broad distribution of metal values is as follows:-

- (i) Generally low values in the sandy shale at the base of the Whyalla Sandstone.
- (ii) Normal copper values are 20-60 ppm; the highest values occur at the top of the black shale, with another zone of high values at the base of the unit (see Fig. 21).

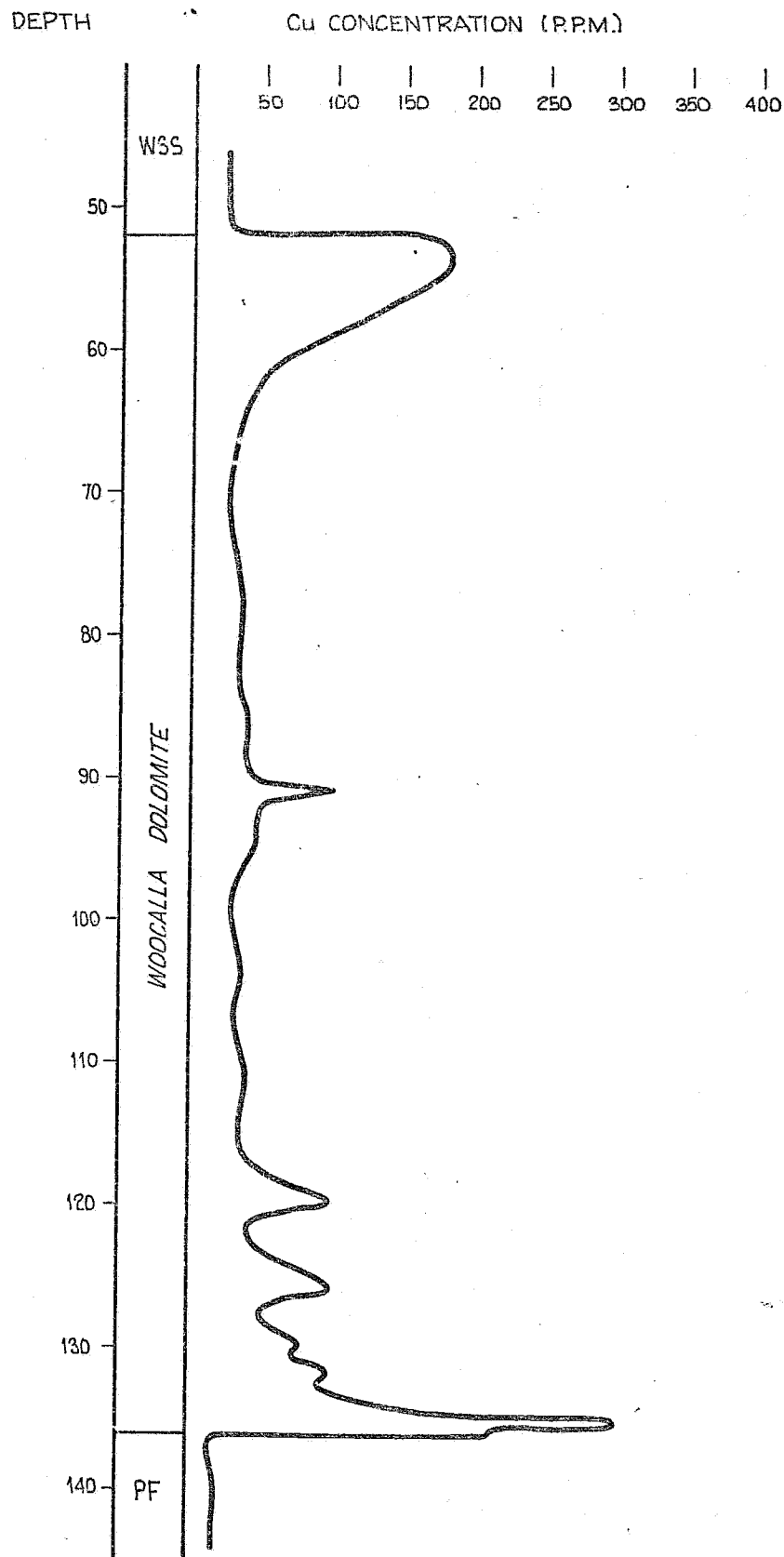


Figure 21. DISTRIBUTION OF Cu VALUES IN BLACK SHALE IN NORANDA'S HOLE LD4.

- (iii) Normal range of Pb values is 50-100 ppm, and for Zn 50-500 ppm; high values are scattered throughout the unit, but a greater concentration of high values is present in the middle to upper portion of the unit.
- (iv) Higher copper values occur towards the edge of the basin of deposition, Pb-Zn values further away from the edge, as shown in Table 2 below.

Hole No.	Estimated Distance from edge of Basin	Max. Metal Values (ppm)		
		Cu	Pb	Zn
LD1	1 km	4600	1600	2200
LD6	2 km	2400	210	320
LD2	4 km	1000	2000	1500
LD5	6 km	1400	1700	1200
LD3	8 km	400	3000	9200
LD4	15 km	180	800	3000

Table 2: Increasing copper concentration towards edge of basin of deposition.

- (v) Since the Woocalla Dolomite undergoes a facies change from black shale to dolomitic shale towards the edge of the basin, high copper values are usually associated with dolomitic shale rather than with black shale (see Fig. 22).

This general pattern is supported by the results of stratigraphic drilling in the Mt. Gunson, Oakden Hills, McLeay and Lake MacFarlane areas. Towards the centre of the basin, Cu and Pb-Zn values are low. Approaching the edge of the basin, Pb-Zn values increase, then decrease while Cu values increase. The increase in copper values is accompanied by a gradual facies change from black shale to dolomitic shale.

North of the Cattlegrid, three rotary drill holes intersected high metal values, indicating a mineralised area of about 3 km². Data from these holes are summarised in Table 3.

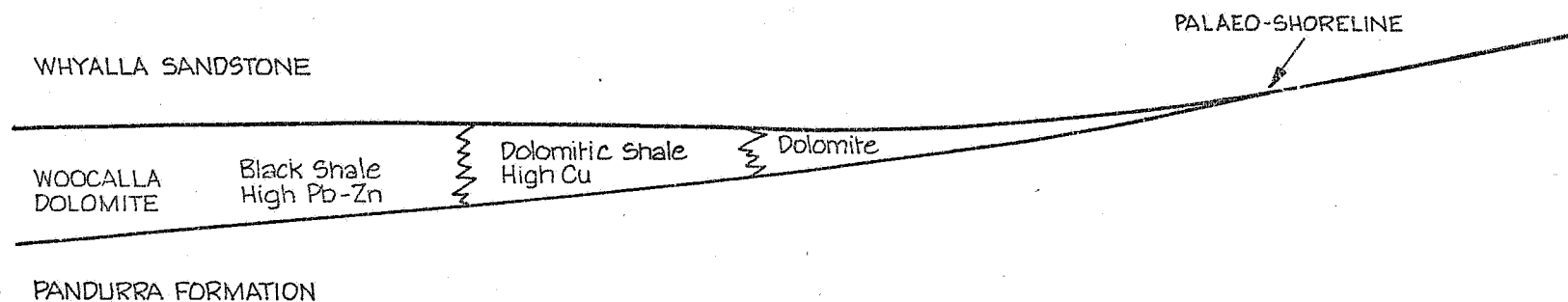


Figure 22. SCHEMATIC DIAGRAM SHOWING HIGH Cu VALUES ASSOCIATED WITH DOLOMITIC SHALE TOWARDS EDGE OF BASIN, Pb-Zn ASSOCIATED WITH BLACK SHALE AWAY FROM THE EDGE

000459

Hole No.	Approx. Distance from Palaeo-Shoreline (km)	Rock Type	Max. Metal Value (ppm)		
			Cu	Pb	Zn
MG21	0.25	Dolomitic Shale	5700	130	820
MG17	0.75	Dolomitic Shale	6500	340	890
MG18	1.25	Black Shale	2400	2400	1100

Table 3: Base metal concentrations in Woocalla Dolomite north of the Cattlegrid. Samples represent 3 m intersections. The sample from hole MG21 was considerably contaminated by sandstone fall-back, so the actual Cu concentration could be in excess of 1%.

In the vicinity of Birthday Siding, hole OK48 intersected black shale with noticeable concentrations of sulphide, (up to 0.5% Cu in a 3 m intersection). The shale here apparently occupies a broad channel in the post-Pandurra palaeo-surface. (see Fig. 23).

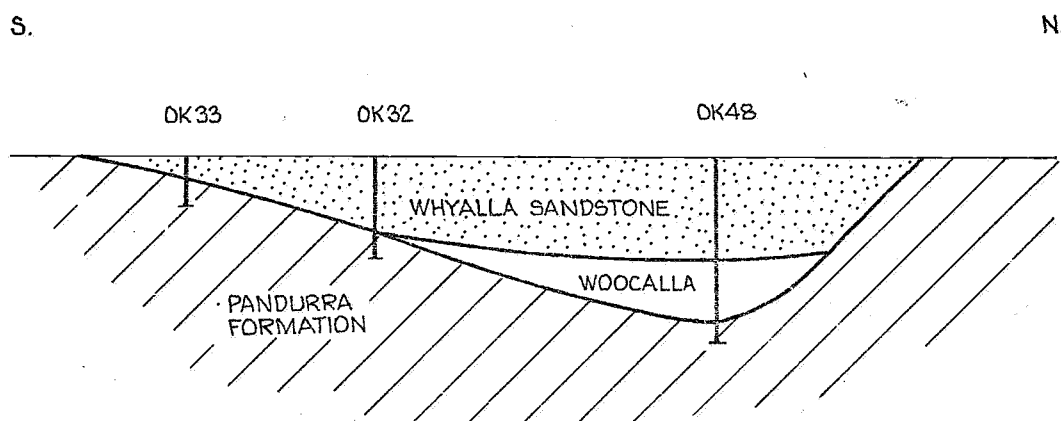


Fig. 23: Black shale occupying a channel in the palaeo-surface near Birthday. (Vertical exaggeration X10).

In the McLeay area, there is a vertical facies change from laminated dolomite up through dolomitic shale into black shale, then into more dolomitic shale with white crystalline dolomite at the top. Copper concentration is at a maximum in the laminated dolomite, decreasing up through the dolomitic shale into the black shale. Above the black shale, copper increases again in dolomitic shale and then drops away in the white crystalline dolomite (see Table 4).

<u>Facies Type</u>	<u>Copper Concentration (ppm)</u>		
	<u>Hole OK16</u>	<u>Hole OK19</u>	<u>Hole OK23</u>
Crystalline Dolomite	60 - 230		
Dolomitic Shale	1000 - 1100	160 - 450	190 - 500
Black Shale	360 - 450	30 - 65	20 - 35
Dolomitic Shale	900 - 1600	300	500
Laminated Dolomite	1000 - 3000	1300	850

Table 4: Cu concentration varying with dolomite facies type in McLeay area.

7.3 Comparison with Other Areas

Bituminous shales throughout the world show abnormally high metal contents. They form in stagnant, non-oxygenated environments which favour the precipitation of sulphides. The Woocalla Dolomite has been compared to the Kupferschiefer of northern Europe by Dunlop and Dunlop (1970), and to the Zambian Copperbelt by Johns (1972).

In the Copperbelt it is suggested (Brandt, 1962; Garlick, 1972) that sulphides formed in lagoons in which reducing conditions prevailed. The lagoons were partly enclosed by basement highs, sand dunes and algal reefs; differential compaction of sediments over the basement highs and the dunes allowed perpetuation of the lagoons. The ore formations are commonly barren over basement highs.

In addition, sulphide distribution in carbonate rocks is apparently related to facies changes. Biostromes form finger-shaped masses around the basement highs. Sulphide occurs in shaley inter-reef rocks rather than in the reef. The sulphides are interpreted as syngenetic deposits produced by reaction of metal ions with hydrogen sulphide. Copper and silver sulphides, being more insoluble, should be precipitated before the more soluble zinc, cobalt and iron sulphide, (Garlick, 1972). From the reef fringe towards progressively deeper water shales, the following horizontal zoning occurs:-

iron oxides ➡ chalcocite ➡ bornite ➡ chalcopyrite ➡ pyrite

A similar situation exists at White Pine, Michigan. White and Wright (1954) consider that the copper is syngenetic, derived from the waters in which the associated sediments were deposited and precipitated in stagnant lagoons enclosed by levees and sand bars in a deltaic environment.

The Kupferschiefer is a bituminous dolomitic shale formed in the Middle Permian Zechstein basin. Although some 20,000 km² of black shale enriched in Cu-Pb-Zn is involved, the areas of bottom stagnation were apparently localised, and the sediments accumulated in a series of shallow nearshore basins (Stanton, 1972). The deposits are frequently zoned vertically, with a copper-rich zone at the base and a lead-zinc zone above, as well as horizontally. The zoning has usually been explained by Garlick's scheme, the copper-rich zone occurring closer to the palaeo-shoreline.

However, in a discussion of Mansfeld-type deposits in Russia, Lur'ye and Gablina (1972) state that the actual distribution of metals does not always coincide with the change in the facies composition of the ore-bearing rocks.

They consider that the main factors governing the localisation of mineralisation are:-

- a) A rapid change from oxidising to reducing conditions (which might coincide with facies change).
- b) Structure of the underlying rocks, where sand-filled river beds may have served as groundwater circulation channels.

7.4 Controls for Mineralisation

The following parameters appear to be important in localising base metal mineralisation in the Woocalla Dolomite:-

7.4.1 Facies Changes - metal sulphides occur in two distinct facies of the dolomite.

- a) Dolomitic mud and silt (Sweet Nell-type), deposited in very shallow water, probably in a supra-tidal environment.
- b) Dolomitic shale, with Cu mineralisation closer inshore and Pb-Zn offshore.

In both cases, the actual controlling factor may be a rapid change from oxidising to reducing conditions as proposed by Lur'ye and Gablina (1972) for the Mansfeld-type deposits.

7.4.2 Depressions in the post-Pandurra palaeo-surface: higher metal concentrations appear to be associated with local depressions (basins or channels) in the post-Pandurra palaeo-surface. In the Winnie Pinnie-Trevenna area, the highest Pb-Zn concentrations are aligned with elongated depressions. At Sweet Nell Mine, the mineralisation occurs in an outlier of shale which was formerly part of a narrow channel, perhaps

with a local depression at Sweet Nell. In the Oakden Hills area, hole OK48 is apparently at the edge of a channel. The mineralisation in the MG14 area appears to be associated with a shallow basin.

7.5 Exploration Techniques

7.5.1 Stratigraphic Drilling

Shallow rotary drilling has been successful in determining the extent of the various rock units, in outlining the distribution of Woocalla facies types, and in indicating zones of mineralisation in the Woocalla Dolomite.

7.5.2 Geochemical Drilling

It seems unlikely that geochemical drilling will outline areas of mineralisation in the Woocalla Dolomite, firstly because the depth of Tregolana Shale is excessive in some areas (30 m in MG18) and secondly because geochemical values in the Whyalla Sandstone are not generally anomalous (see Table 5).

<u>Hole</u> <u>No.</u>	<u>Max. Cu</u> <u>Conc. %</u>	<u>Geochemical Value (ppm)</u>					
		<u>Top of Whyalla Ss.</u>			<u>At Water Table</u>		
		<u>Cu</u>	<u>Pb</u>	<u>Zn</u>	<u>Cu</u>	<u>Pb</u>	<u>Zn</u>
MG14	1.83	15	50	10	30	100	30
MG14D	2.53	10	20	10	50	20	15
MG17	0.65	30	25	80	20	20	40
MG18	0.24	10	35	30	10	35	30
MG21	0.57	65	20	25	60	20	20
OK11	0.43	15	40	30	10	40	25

Table 5: Geochemical values in Whyalla Sandstone above mineralisation in Woocalla Dolomite.

7.5.3 Geophysics

The MG14 area was delineated as a target by the INPUT survey (Langron, 1973), which also outlined Anomaly "C" in the Powerline area. Subsequent drilling near Anomaly "C" (hole PL1) showed that dolomitic mud fills what appears to be a depression in the post-Pandurra palaeo-surface. This material may be mineralised but assays are not yet available.

Follow-up geophysics in the MG14 area produced a weak E.M. cross-over and a quite good I.P. anomaly coinciding with the INPUT anomaly.

7.5.4 Conclusion

The INPUT method is possibly picking up depressions in the post-Pandurra palaeo-surface. If so, this method, accompanied by rotary drilling should be most useful in delineating those areas of Woocalla Dolomite which satisfy the requirements of facies type and relation to depressions in the palaeo-surface.

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